

Unlocking the Capabilities of Thought: A Reasoning Boundary Framework to Quantify and Optimize Chain-of-Thought

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Abstract

Chain-of-Thought (CoT) reasoning has emerged as a promising approach for enhancing the performance of large language models (LLMs) on complex reasoning tasks. Recently, a series of studies attempt to explain the mechanisms underlying CoT, aiming to deepen the understanding of its efficacy. Nevertheless, the existing research faces two major challenges: (1) *a lack of quantitative metrics to assess CoT capabilities* and (2) *a dearth of guidance on optimizing CoT performance*. Motivated by this, in this work, we introduce a novel reasoning boundary framework (RBF) to address these challenges. To solve the lack of quantification, we first define a reasoning boundary (RB) to quantify the upper-bound of CoT and establish a combination law for RB, enabling a practical quantitative approach applicable to various real-world CoT tasks. To address the lack of optimization, we propose three categories of RBs. We further optimize these categories with combination laws focused on RB promotion and reasoning path optimization for CoT improvement. Through extensive experiments on 27 models and 5 tasks, the study validates the existence and rationality of the proposed framework. Furthermore, it explains the effectiveness of 10 CoT strategies and guides optimization from two perspectives. We hope this work can provide a comprehensive understanding of the boundaries and optimization strategies for reasoning in LLMs. Our code and data are available at <https://github.com/LightChen233/reasoning-boundary>.

1 Introduction

In recent years, Large Language Models (LLMs) have demonstrated increasing capabilities and applications across various tasks [Zhao et al., 2023, Chang et al., 2023, Pan et al., 2023, Qin et al., 2024a]. Notably, advanced LLMs, such as GPT [Brown et al., 2020, OpenAI, 2022, 2023], PaLM [Anil et al., 2023] and LLaMa [Touvron et al., 2023a,b, Meta, 2024] series have demonstrated emergent capabilities, particularly like Chain-of-Thought (CoT) [Nye et al., 2022, Wei et al., 2022]. This methodology enables models to verbalize step-by-step reasoning, thereby enhancing prediction accuracy by basing decisions on the logical rationale [Wei et al., 2022, Kojima et al., 2022, Hu et al., 2024, Qin et al., 2023, Zhuang et al., 2023, Chen et al., 2024a].

Recently, some research in the literature has begun to investigate the mechanism of CoT to enhance the understanding of its operational nature. To this end, Madaan et al. [2023] and Wang et al. [2023a]

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first give a qualitative boundary conclusion through a large number of experiments on the natural language planning capability: The CoT is limited by the reasoning logic in the context demonstrations. Bi et al. [2024] investigate these boundaries on the code planning capability, by training LLMs on CoT samples of varying difficulties. It demonstrates LLMs are unable to learn or effectively manage tasks that exceed a certain complexity upper-bound. To delve deeper into potential constraints of CoT, Feng et al. [2024] develop a theoretical framework on the single-step calculation capability, suggesting that there is an upper-bound of model performance dependent on the length of input in single-step reasoning processes. Although existing research has made some progress, where the boundaries of CoT lie and how these boundaries affect the performance of CoT are still unresolved questions. Specifically, the existing work still faces two major challenges:

- **Lacking quantification metrics for CoT:** Current research primarily relies on qualitative assessments of CoT performance, which leads to the absence of quantitative metrics. It hinders the ability to objectively compare different CoT approaches and establish a definitive upper-bound for CoT capabilities.
- **Lacking optimization guidance for CoT:** While current research prioritizes understanding the mechanisms underlying CoT reasoning, there is a dearth of guidance on optimizing CoT performance. This gap hinders the transformation of CoT research into actionable strategies for enhancing model capabilities.

Motivated by this, in this work, we introduce a reasoning boundary framework (RBF) to thoroughly examine and optimize the boundaries of current LLMs. Specifically, to address the quantification challenge, we propose a new concept, named reasoning boundary (RB) to quantify the upper-bound on task-specific reasoning complexity within a model. Furthermore, to explore more practical scenarios, we present the combination law of RBs to generalize the RB for quantification in more real and complex scenarios. To address the CoT optimization challenge, we propose and analyze three reasoning boundary intervals, guiding optimization through improved RB and optimized reasoning paths based on the combination law, which achieves state-of-the-art performance in our proposed benchmark. We extensively validate the efficacy of our framework across 27 models and 5 tasks: arithmetic computing, mathematical reasoning, multi-hop question answering, and multilingual mathematical reasoning.

Our main contributions are as follows:

- To the best of our knowledge, this is the first work to propose a reasoning boundary framework (RBF) to quantify the upper-bound of CoT. Furthermore, we establish the combination law of RB as the weighted harmonic mean of fundamental RBs to address practical CoT tasks.
- To solve the lack of CoT optimization, we define three categories of RBs. Based on the combination law and the nature of these RBs, we effectively improve the existing CoT strategies by RB promotion and reasoning path optimization.
- We validate the existence and rationality of our framework on 27 models and 5 CoT tasks. Furthermore, we explain the optimal performance from two optimization perspectives in numerous CoT strategies. We consider both optimal perspectives and propose a minimum acceptable reasoning path (MARF) prompting to achieve state-of-the-art performance.

2 Quantification Methodology

2.1 Reasoning Boundary

In order to quantify the capacity for complex reasoning in LLMs, we introduce an upper-bound concept termed reasoning boundary (RB), which formally defines the degree of ease that an LLM can handle within a specific reasoning process. In simpler terms, as shown in Figure 1 (a), RB reflects the limit beyond which a model’s accuracy significantly degrades. Mathematically, RB is defined for a model m and a task t as the maximum of problem difficulty d at which the model’s accuracy reaches a predefined threshold K_1 :

$$\mathcal{B}_{Acc=K_1}(t|m) = \sup_d \{d | Acc(t|d, m) = K_1\}, \quad (1)$$

where $Acc(t|d, m)$ represents the accuracy of the model’s accuracy on task t with difficulty d . Difficulty can be measured by factors like the number of reasoning steps or computational complexity. For brevity, we denote RB as $\mathcal{B}(t|m)$ in subsequent sections.

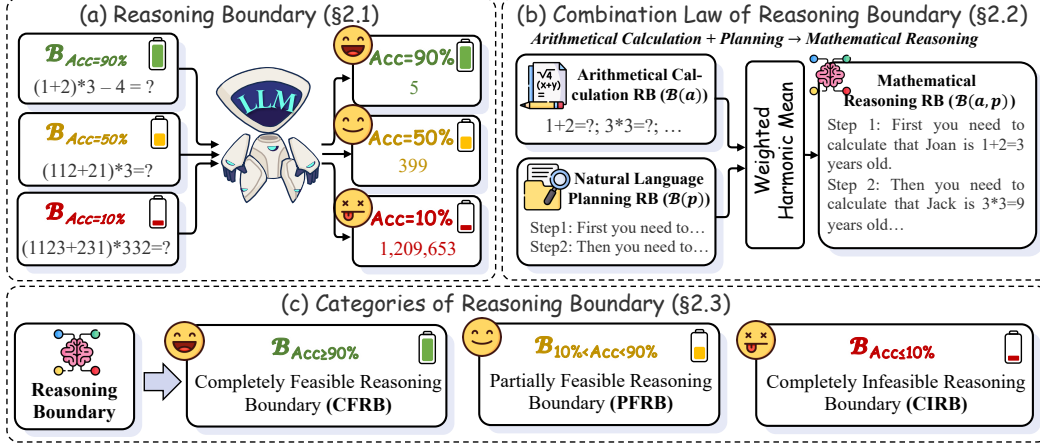


Figure 1: Overview of the introduced concepts.

Conclusion: The reasoning boundary for a model is defined by its ability to achieve a specific accuracy for a given task difficulty.

2.2 Combination Law of Reasoning Boundary

In practical scenarios, models often require the integration of multiple capabilities to address a single task effectively. To quantify how a large language model can be boosted by the cooperation of multiple capabilities through the CoT mechanism, we introduce the “*Combination Law of RB*”, giving a concrete formula of the upper-bound of the CoT. The law estimates the unified reasoning boundary $B_{Acc=K_1}(t_1, t_2, \dots, t_n|m)$ for n tasks within a model m , which is formulated as:

$$B_{Acc=K_1}(t_1, t_2, \dots, t_n|m) \approx \frac{1}{(n-1) \sum_{i=1}^n \frac{N_i}{B_{Acc=K_1}(t_i|m) - b_i}}, \quad (2)$$

where $B_{Acc=K_1}(t_i|m)$ denotes the reasoning boundary of model m for task t_i . N_i , and b_i are scaling factors, which are only affected by the related task. As shown in Figure 1 (b), Equation (2) provides a mathematical formula to estimate the combined RBs from the independent ones, enabling deeper insights into model behavior for intricate tasks. See Appendix A for detailed mathematical analysis.

Furthermore, the combination law for reasoning boundary demonstrates favorable theoretical properties, with broad applicability across diverse scenarios and flexibility in accommodating various boundary segmentation methods. For detailed practical application, please refer to Appendix B.

Conclusion: The combination law of reasoning boundary satisfies the weighted harmonic average of each basic reasoning boundary.

2.3 Categories of Reasoning Boundary

Furthermore, in order to guide the optimization of CoT and more convenient expression, as shown in Figure 1 (c), we define the following three categories of RBs based on their empirical accuracy:

Completely Feasible Reasoning Boundary: We define that the part with an accuracy greater than 90% is a completely feasible reasoning boundary ($CFRB = B_{Acc \geq 90\%}(t_1, t_2, \dots, t_n|m)$), which means that LLMs can effectively grasp the performance of this part.

Completely Infeasible Reasoning Boundary: We believe that the part with an accuracy less than 10% is a completely infeasible reasoning boundary ($CIRB = B_{Acc \leq 10\%}(t_1, t_2, \dots, t_n|m)$), which means that the model can never effectively grasp the performance of this part.

Partially Feasible Reasoning Boundary: We define the RB in the rest part except CFRB and CIRB as a partially feasible reasoning boundary ($PFRB = B_{10\% < Acc < 90\%}(t_1, t_2, \dots, t_n|m)$), which requires the model to repeat thinking or more clear information to solve the problem.

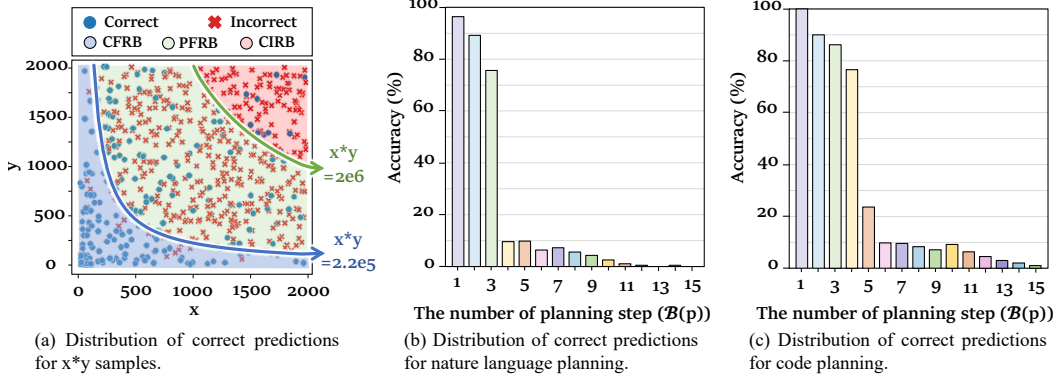


Figure 2: Existence Verification for Reasoning Boundary.

We analyze the nature of these three categories of RB in detail (in Section 4.3), and further utilize the combination law to optimize these three reasoning boundaries (in Section 5), so as to provide effective suggestions and guidance to support future CoT optimization.

3 Experimental Setup

Benchmark Settings To assess the reasoning boundaries of LLMs, we require a dataset rich in RB. This necessitates tasks with evenly distributed complexities and reasoning steps that challenge the models’ upper-bounds. To meet these requirements, we introduce BIGGSM, a new dataset offering greater calculation complexity and longer reasoning chains. The detailed construction process for BIGGSM is provided in Appendix C.

Model Settings Except for model expansion experiments, all experiments are conducted on GPT-3.5-Turbo. Following the setting of Wei et al. [2022], in our CoT experiment, all multi-step reasoning tasks utilize three manually constructed demonstrations. In addition, for all the experiments, top-p is selected from $\{0.95, 1\}$. Temperature is selected from $[0, 1]$ and serves as the main error variable.

4 Empirical Analysis of Reasoning Boundary

4.1 Existence Verification for Reasoning Boundary

In this study, we investigate the hypothesis that an LLM exhibits varying levels of reasoning boundary across various tasks. To this end, we will verify whether the model has widespread reasoning boundary in various tasks in the following three tasks:

Basic Arithmetic Calculation First, to investigate the existence of RB, we first examine basic arithmetic operations (including addition, subtraction, multiplication, and division). As illustrated in Figure 2 (a), the results reveal significant performance variations across three distinct regions. For multiplication, accuracy surpasses 90% for results up to $2.2e5$. Conversely, accuracy falls below 10% for products exceeding $2e6$. Similar presences of varying RBs are observed for other operations, which verifies the existence of reasoning boundary in basic arithmetic calculation tasks. Further results and implementation details are provided in Appendix D.

Nature Language Planning We further investigate RB in natural language planning tasks for mathematical reasoning. We prompt the model to generate plans and assess their accuracy through manual evaluation. There is a strong correlation between the number of reasoning steps and LLMs’ performance in Figure 2 (b). When the model meets the question with fewer than 2 reasoning steps, accuracy surpasses 90%. Conversely, when reasoning steps exceed 4, accuracy falls below 10%. This finding suggests that there are also three different RB categories in natural language planning tasks.

Code Planning For further extensive exploration, we further prompt LLMs by PAL [Gao et al., 2023] to generate code-format plans and evaluate them by manual annotation. As shown in Figure 2 (c), the code planning task is similar to natural language planning, which is also an obvious division and different categories of RBs. Notably, since code planning utilizes code for clearer logic and reduced expression complexity, its planning accuracy surpasses that of natural language planning.

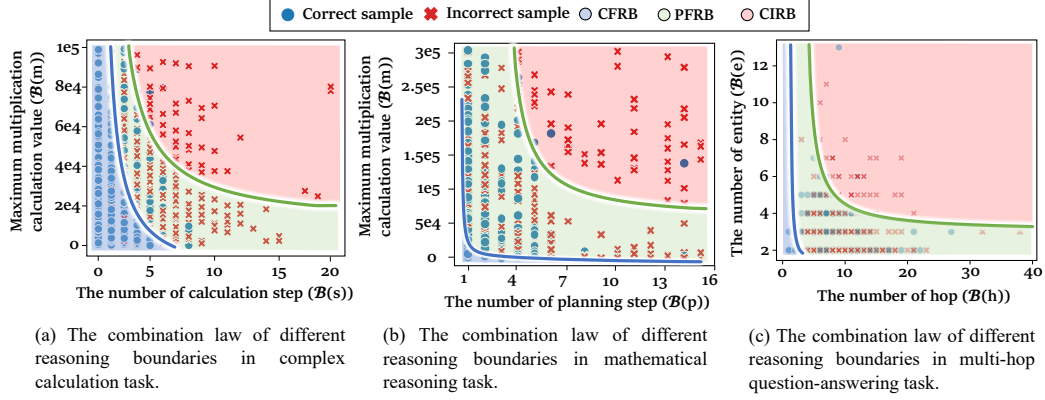


Figure 3: Combination law verification of RB on different tasks. More verification results on other tasks are shown in Figure 12.

4.2 Combination Law Verification on Different Tasks

Combination Law in Complex Arithmetic Calculation Building on the proof of Equation (13), we hypothesize that the combination law for RB in the complex arithmetic calculation is the harmonic average of the arithmetic calculation RB and calculation planning RB. To verify this, we designed an experiment focusing on formulas containing addition, subtraction, and multiplication, like “ $(1 + 2) * 3 - 4$ ”. Since addition and subtraction complexities are assumed to be around $1e15$ (as shown in Figure 13), the arithmetic calculation RB primarily depends on the multiplication RB and calculation planning RB. Therefore, as shown in Figure 3 (a), there are two obvious RB lines, namely $\mathcal{B}_{Acc=90\%}$ and $\mathcal{B}_{Acc=10\%}$, which are completely consistent with the combination law of these basic RB based on the Equation (2). Besides, these two lines also clearly divide the RBs into three categories.

Combination Law in Mathematical Reasoning Inspired by Tan [2023b], Xiao and Liu [2024], we posit that the natural language mathematical CoT task is determined by two sub-tasks: step planning task and step calculation task for global logic planning and local mathematical calculation. Furthermore, each model output step requires a single basic operation, resulting in a step calculation boundary close to the maximum number of multiplications, denoted by $\mathcal{B}(c) \approx \mathcal{B}(m)$. Formally, with step planning RB denoted by $(\mathcal{B}(p))$ and the step calculation RB by $(\mathcal{B}(c))$, then the combined RB satisfies the following law:

$$\mathcal{B}^{CoT}(c, p) = \frac{1}{\frac{N_1}{(\mathcal{B}(c)-b_1)} + \frac{N_2}{(\mathcal{B}(p)-b_2)}}. \quad (3)$$

As illustrated in Figure 3 (b), the actual performance distribution of RB (including $\mathcal{B}_{Acc=90\%}$ and $\mathcal{B}_{Acc=10\%}$) in natural language mathematical reasoning task fully aligns with the proposed combination law in Equation (3). Additionally, there are also obviously three RBs in Figure 3 (b).

Combination Law in Multi-hop Reasoning Beyond the realm of mathematics, we further extend our exploration of the combination law to the field of multi-hop question answering. Specifically, we validate our law on HotpotQA [Yang et al., 2018], where we define the reasoning boundary as the combination of global hop-planning RB and local knowledge entity reasoning RB. As shown in Figure 3 (c), $\mathcal{B}_{Acc=90\%}$ and $\mathcal{B}_{Acc=10\%}$ also satisfy the weighted harmonic mean of these two sub-reasoning boundaries. It is also proved that, in addition to math-related tasks, multi-hop question answering also satisfies our proposed combined law and also exhibits three distinct RBs. We will describe in detail how to calculate the combination law on multi-hop reasoning in Appendix E.

4.3 Nature Analysis for different Reasoning Boundary

According to the definition of different RBs, we have divided the problem into three parts for LLMs. In this section, we will verify whether the defined RB adheres to the intrinsic nature of the model itself. We will discuss the natures of these RBs in detail:

CFRB means complete mastery of the model even without demonstration. According to the definition, we assume that a question within CFRB implies a comprehensive understanding of the associated issue for a certain LLM. To verify this, following Zhang et al. [2022] and Wei et al. [2022], we formulate a mathematical request and generate chain-of-thought rationale and answer through

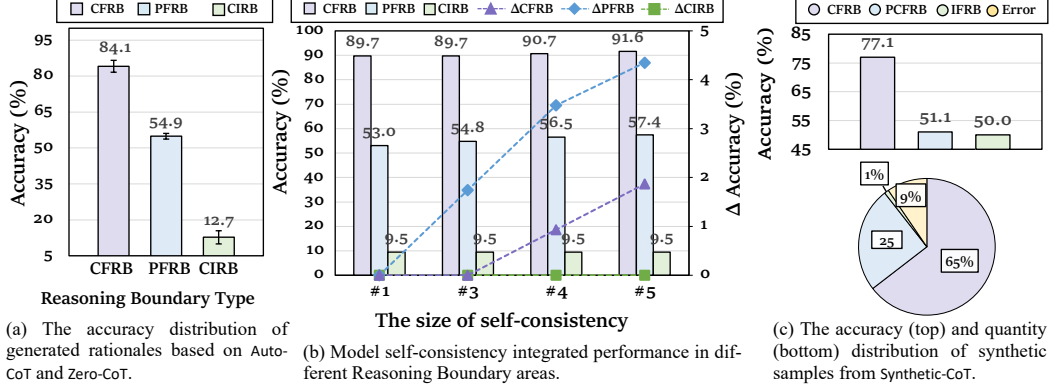


Figure 4: Nature analysis at different reasoning boundaries.

zero-shot prompting without any demonstration. As shown in Figure 4 (a), it still achieves 29.2% improvement in CFRB on generating the correct rationale compared to other RBs. This also proves that the model can indeed master tasks well on the questions in CFRB.

PFRB means moderate confidence in its solution and needs consensus building process. To gauge the level of performance and confidence, we draw parallels to human decision-making, where moderate confidence often necessitates multiple times of consensus building. Inspired by this, we investigate it on Self-Consistency [Wang et al., 2022], which integrates results from various reasoning answers to reach a conclusive answer. Figure 4 (b) demonstrates that as the integration of reasoning paths increases, the accuracy improves significantly within PFRB compared with other RBs. This suggests that within PFRB, the LLM exhibits moderate confidence in solving problems, which needs multiple consensus building.

CIRB exhibits poor reasoning performance even with consensus building. As illustrated in Figure 4 (a), questions in CIRB display extremely low accuracy (around 9.5%). And the model shows consistently poor performance and no improvement on Self-consistency in this boundary in Figure 4. It signifies that the model exhibits poor reasoning performance.

LLM has self-awareness of its own RBs. In parallel, a natural question arises: *Is the model capable of discerning its inherent RBs?* To investigate this, we employ the Synthetic-CoT [Shao et al., 2023] to prompt LLM to generate CoT data. As depicted in Figure 4 (c), the results demonstrated that there are over 65% of generated samples within CFRB, which achieves a much higher percentage and performance than other RBs. This suggests that LLMs possess an intrinsic understanding of their RBs and constraints to generate the task they grasp, indicative of a potential for self-assessment.

Takeaways: (1) Reasoning boundary (RB) and the combination law of RB are both widespread across a series of tasks. (2) Different categories of RB can reflect the corresponding performance, and the model can also have a self-understanding of its own RB.

5 RB-based CoT Optimization

5.1 How can we improve CoT by optimizing RB?

Based on our framework, the reasoning boundary limits the performance of the model. The simplest approach to improve CoT is to optimize the step calculation RB $\mathcal{B}(c)$ to promote the value of RB. Specifically, Tool-Usage [Paranjape et al., 2023] and Program-of-Thought (PoT) [Chen et al., 2024b] have shown significant success in CoT optimization. We explain the rationale behind their effectiveness, why PoT consistently outperforms direct Tool Usage [Yao et al., 2023, Chen et al., 2023], and take them as examples to demonstrate how to improve CoT by promoting RB.

Tool Usage can boost the value of RB for an LLM. When the model uses tools [Paranjape et al., 2023], we can simply think that the model can perform calculations with infinite precision, so that the RB of mathematical calculations tends to infinity, viz $\mathcal{B}(c) \rightarrow +\infty$. It is obvious that the combined

Model	BIGGSM		
	Acc. ($\uparrow$)	Input Token ($\downarrow$)	Output Token ($\downarrow$)
CoT	57.00 ± 0.93	780.43	96.76 ± 3.22
RB-Optimized Methods			
Tool Usage	71.64 ± 0.66	688.43	129.53 ± 3.82
PoT	78.25 ± 1.09	657.43	78.25 ± 1.09
Reasoning-Path-Optimized Methods			
Least-to-most	58.25 ± 3.28	679.59	176.09 ± 15.22
Complex-CoT	59.78 ± 0.60	1111.43	131.82 ± 1.91
CoT+MARP	64.37 ± 2.24	614.43	95.12 ± 0.77
PoT+MARP	80.55 ± 2.40	576.43	76.34 ± 2.84

Table 1: Main experimental results on GPT-3.5-Turbo. Results on different benchmarks are shown in Table 2.

RB of the model can be calculated as:

$$\mathcal{B}^{\text{Tool}}(c, p) = \lim_{\mathcal{B}(c) \rightarrow +\infty} \frac{1}{\frac{N_1}{(\mathcal{B}(c) - b_1)} + \frac{N_2}{(\mathcal{B}(p) - b_2)}} = \frac{\mathcal{B}(p) - b_2}{N_2}. \quad (4)$$

Easy to get, $\mathcal{B}^{\text{Tool}}(c, p) > \mathcal{B}^{\text{CoT}}(c, p)$, this shows that Tool Usage can improve the boundary of reasoning. This explains why Tool Usage can have better performance than vanilla CoT (as shown in Table 1). Furthermore, as shown in Figure 5, the distribution of theoretical RB and the actual one almost perfectly coincide. This also demonstrates the reliability and applicability of our theory.

Program-of-Thought can further enhance the value of LLM’s RB. Equation (4) reveals that an LLM’s RB hinges entirely on its planning capability. Since natural language can be verbose, it hinders the planning capability of LLM [Gao et al., 2023, Hu et al., 2023, Puerto et al., 2024, Chen et al., 2024b]. PoT [Chen et al., 2023] offers a clearer representation of logic using code, allowing for clearer planning (as shown in Figure 2 (b, c)). This leads to finer-grained planning reasoning $\mathcal{B}^*(p) > \mathcal{B}(p)$. Then the PoT reasoning boundary $\mathcal{B}^{\text{PoT}}(c, p) > \mathcal{B}^{\text{Tool}}(c, p)$, aligning with the observed performance gains of PoT over Tool Usage (see Table 1). Furthermore, Figure 5 visually demonstrates that PoT’s theoretical and practical reasoning boundaries consistently outperform Tool Usage. This reinforces the theoretical advantage of PoT and its empirical effectiveness.

5.2 How can we improve CoT based on a certain RB?

Enhancing RB is crucial for optimizing CoT, but requires changes to the model or its reasoning architecture to be effective. Therefore, we need to consider how to optimize the reasoning path so that the difficulty satisfies the RB ($d^* = \mathcal{B}_{\text{Acc}=K_1}$) instead of the original RB ($d = \mathcal{B}_{\text{Acc}=K_2}$), where $K_2 < K_1$. According to Equation (3), $\mathcal{B}$ is affected by both arithmetical RB and planning RB. Given $\mathcal{B}$, we consider optimizing reasoning ability from the following two strategies as examples ²:

Complex CoT (CCoT): By increasing the boundary of planning to reduce the pressure of single-step calculation, reduce the arithmetical RB, and then get smaller d ; However, it introduces more planning steps, which adds the planning pressure. As shown in Figure 6, the model performance first increases and then decreases with the increasing number of CCoT steps.

Least-to-Most (LtM): By dividing multiple sub-questions to reduce the pressure of local planning within a sub-question, reduce the boundary of local planning, and then get smaller d . However, even though it can release local planning pressure (as demonstrated in Figure 7), this approach simultaneously intensifies global planning pressure by generating an excessive number of sub-questions (as depicted in Figure 15).

Limitation: (1) CCoT needs to keep balance in the number of reasoning steps and calculation pressure. (2) Although the pressure of local planning has been reduced, LtM has not effectively reduced the pressure of global planning, nor the pressure of optimization calculations.

Minimum acceptable reasoning paths prompting can further achieve better CoT within a specific RB. To address the aforementioned two issues, we proposed Minimum Acceptable Reasoning

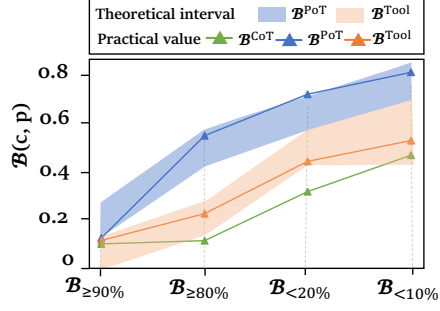


Figure 5: Analysis of the impact of Tool Usage and PoT on reasoning boundary $\mathcal{B}(c, p)$.

²See detailed analysis for these two strategies in Appendix F.

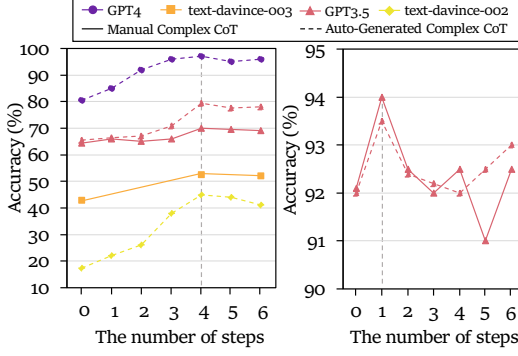


Figure 6: Correlation between the number of steps and performance of Complex-CoT on GSM8K (left) and SingleEq (right). See Appendix G for more meta-analysis results.

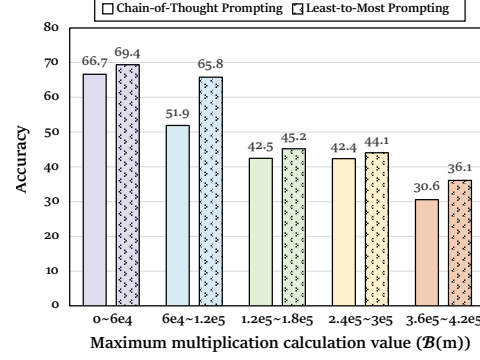


Figure 7: The performance distribution of Least-to-Most prompting on different calculation amounts.

Paths (MARPs). Our first objective is to alleviate the computational burden of the model. We achieve this by introducing instructions that set an upper limit on its single-step computational capacity, thereby optimizing the boundary of its computational reasoning. Secondly, we aim to enhance the model’s acceptability. Within the calculation and planning boundary, we increase the amount of computation performed in each step in demonstrations as much as possible while simultaneously reducing the number of global planning steps, which effectively mitigates planning pressure. As shown in Table 1, MARP demonstrably improves model performance and effectively reduces the token consumption. By maximizing operations per step, MARP leads to a more streamlined and efficient problem-solving process. Detailed descriptions of this strategy are shown in Appendix G.3.

Takeaways: (1) Tool-Usage and PoT can be utilized to optimize CoT by the calculation and planning reasoning boundary optimization. (2) MARP can well lessen planning and calculation pressure by problem optimization in certain RB (3) Users can effectively optimize CoT performance by optimizing the reasoning boundary and the problem.

6 Expansion Verification & Exploration

RB can be extended to various models. To extend our mechanism’s applicability, we verify the mechanism on 25 diverse models (details in Table 3). As shown in Figure 8 (a), we observe a positive correlation between reasoning boundary and model accuracy on mathematical benchmarks. Moreover, the models that use mathematical data such as MathInstruct for SFT, often have interesting outliers that are different from the general LLMs’ area, but they also satisfy a positive correlation with

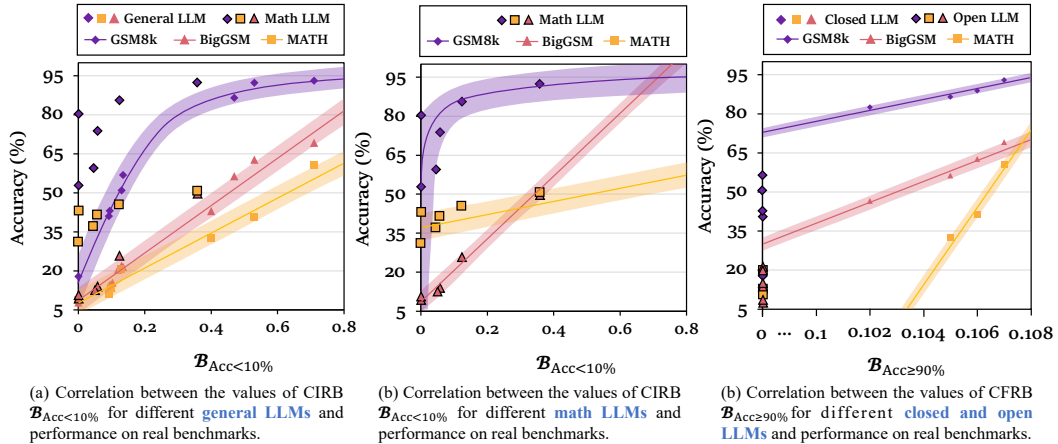


Figure 8: Correlation between the values of RB for different models and performance on real benchmarks. See Appendix H for more empirical details.

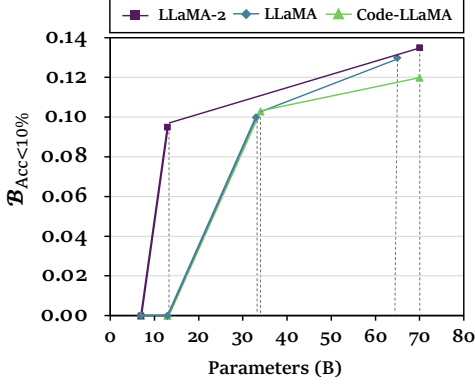


Figure 9: Scaling law correlation between model parameters and CIRB.

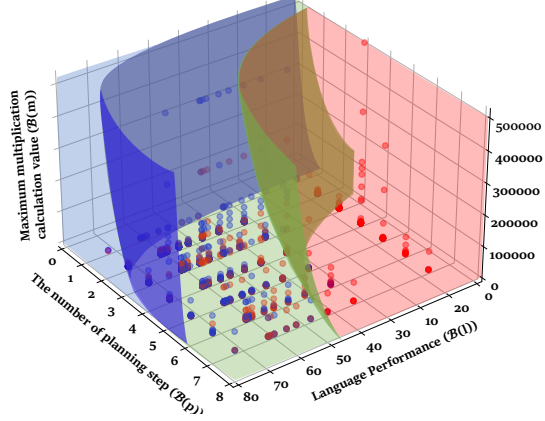


Figure 10: Different boundaries on MGSM.

our RBs (as shown in Figure 8 (b)), which helps determine if the model underwent mathematically targeted training.

However, as shown in Figure 8 (c), we find some interesting phenomena. For example, the main difference between the current open-source model and the closed-source model is still CFRB. Except for the closed source model, the CFRB of all models is 0. It shows the potential and the direction of the model optimization. Furthermore, a scaling law of RB can also emerge (as shown in Figure 9): reasoning boundary increased with model parameter count and data quality.

RB can be extended to more tasks. To assess the RB in more tasks, we evaluate them on a multilingual mathematical reasoning task. Inspired by Qin et al. [2024b], we hypothesize that multilingual RB, assessed through direct answer accuracy across different languages, mathematical computation RB, represented by the maximum product result, and reasoning planning RB, indicated by the planning steps, are orthogonal dimensions of performance. We propose that these RBs can be effectively combined using a weighted harmonic mean. As illustrated in Figure 10 confirms that the combined RB maintains the expected three different RBs. Detailed implementation description is shown in Appendix I.

7 Related Work

In this section, we review recent literature related to Chain-of-Thought (CoT) prompting, focusing on theoretical and empirical investigations. Madaan et al. [2023], Wang et al. [2023a], Saparov and He [2023], He-Yueya et al. [2023], Zhang et al. [2024], Wang et al. [2024] and Prystawski et al. [2024] qualitatively show that the LLMs learn the reasoning chain based on the demonstrations in the context. Besides, Lampinen et al. [2022] and Tan [2023a] find a causal link between generated intermediate steps and the final answers during a series of qualitative experiments. Wang et al. [2023c], Hanna et al. [2024] and Dutta et al. [2024] study neural substructure within the LLMs, embodying CoT reasoning from a white-box mechanism perspective, demonstrating that LLMs deploy multiple parallel answer generation paths internally.

Recently, a large amount of work has demonstrated the upper-bounds and limitations of LLM in various CoT tasks [Qin et al., 2023, Imani et al., 2023, Huang et al., 2024, Sprague et al., 2024]. Bi et al. [2024] investigate these bounds on planning capability in code generation by training LLM on CoT samples of varying difficulties. Their findings suggest that LLMs have a limited capacity to learn or manage tasks exceeding a certain complexity threshold. Further understanding of the CoT upper-bound, Merrill and Sabharwal [2023], Li et al. [2023] and Feng et al. [2024] analyze single-step arithmetic capability, which suggests an upper bound on model performance related to input length in single-step reasoning processes.

Despite advancements in CoT explanation for LLMs, significant challenges remain, including the absence of quantifiable metrics for CoT’s upper-bounds and the deficiency in optimization guidelines. To tackle this, we propose a reasoning boundaries framework (RBF) to systematically quantify and optimize various CoT approaches. This framework offers a transferable and user-friendly

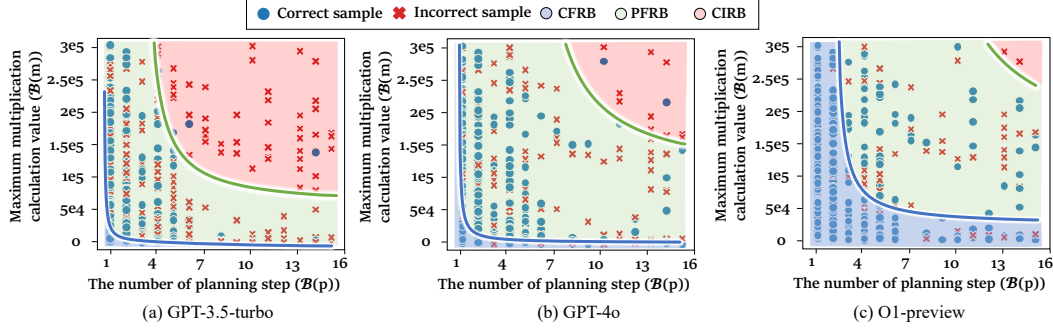


Figure 11: Combination law verification of reasoning boundaries on GPT-series models.

methodology to enhance model performance from a mechanistic perspective. We anticipate that it will furnish systematic insights for ongoing research and inform future developments in the field.

8 Discussion

Discussion on the Boundaries Improvements Furthermore, in order to better understand the best existing LLMs, we utilize RBF to test the current most advanced GPT-series models. As shown in Figure 11, all reasoning boundaries improve a lot compared to the last version which also achieves performance enhancement. Notably, the CFRB increases slightly compared with the improvement of CIRB between GPT-3.5 and GPT-4o. But o1 significantly improves the CFRB. Furthermore, as shown in Figure 14 in Appendix, o1 shows extremely significant improvements on CFRB, which is almost three times of other models. We attribute it to the fact that the advanced Reinforce-Learning and Inference Scaling strategies play a key role in improving this part of the ability compared with the normal improvements in CFRB, which might trigger more in-depth research.

Broader impacts. Our framework is the first work to quantify the reasoning upper-bound of LLMs. This enables the explanation for a huge part of the valid CoT framework. We hope that our work can provide new insights and more systematic guidance for future interpretability analysis of CoT. For social impact, this work may have a certain impact on the controllable and explainable AGI.

Limitations & Future. Due to the cost and time constraints, this work does not discuss the complex relationships such as causal conditions among the basic RBs. In addition, evaluating the robustness and applicability of CoT reasoning boundaries-related techniques in dynamic scenarios will be crucial for future research.

9 Conclusion

This study introduces a novel reasoning boundaries framework (RBF) to quantify and optimize the limitations of LLMs in CoT tasks. Specifically, we propose the concept of reasoning boundaries (RBs) and the combination law of RBs in more complex scenarios for quantitative metrics. We further introduce three categories of RB for CoT optimizations. The framework is validated through extensive experiments across 27 models and 5 tasks. Furthermore, we improve the CoT in both RB and question optimization perspectives to achieve state-of-the-art performance in BIGGSM. We hope that this framework paves the way for further research on understanding and enhancing LLMs' reasoning capabilities.

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Appendix

A Mathematical Analysis & Proof

A.1 Definitions & Assumptions

In order to further quantify and analyze the combination law of RB, we will make the following definitions and assumptions about the properties of RB:

Definition 1 *If a basic RB in the combined RB takes infinity, then the combined RB only depends on the remaining basic RBs.*

That is, the model is no longer limited by a certain ability when solving tasks, and only needs to focus on other ability shortcomings. Formally, combination law satisfies that³:

$$\mathcal{B}(t_1, t_2, \dots, t_n | m) = \mathcal{B}(+\infty, t_2, \dots, t_n | m) + \mathcal{B}(t_1, +\infty, \dots, t_n | m) + \dots + \mathcal{B}(t_1, t_2, \dots, +\infty | m) \quad (5)$$

$$= \mathcal{B}(t_2, t_3, \dots, t_n | m) + \mathcal{B}(t_1, t_3, \dots, t_n | m) + \dots + \mathcal{B}(t_1, t_2, \dots, t_{n-1} | m) \quad (6)$$

$$= (n-1) \sum_{i=1}^n \mathcal{B}(+\infty, \dots, +\infty, t_i, +\infty, \dots, +\infty | m) \quad (7)$$

Definition 2 *If all basic RBs are infinite, it means that the model is omnipotent, and the combined RB is also infinite.*

Formally, the combination law satisfies that:

$$\mathcal{B}(+\infty, +\infty, \dots, +\infty | m) = +\infty \quad (8)$$

Assumption 3 *The combination law function is continuously differentiable everywhere.*

Assumption 4 *All basic reasoning boundary for combined reasoning boundary are mutually independent.*

A.2 The Proof of Combination Law

Based on the above definitions and assumptions, we need to prove that the combination law is a combined RB and is the weighted harmonic average of two basic RBs.

Proof. Since Taylor expansion cannot be performed on infinity and the function needs to converge, we set $t_i = \frac{1}{x_i} + b_i$ and $\frac{1}{\mathcal{B}(t_1, t_2, \dots, t_n | m)} = \mathcal{B}^*(x_1, x_2, \dots, x_n | m)$. Then following Equation (7), we can get the $\mathcal{B}^*(x_1, x_2, \dots, x_n | m)$ as:

$$\mathcal{B}^*(x_1, x_2, \dots, x_n | m) = (n-1) \sum_{i=1}^n \mathcal{B}^*(0, \dots, x_i, \dots, 0 | m). \quad (9)$$

According to the Taylor expansion formula, we expand this formula at $x_i \rightarrow k_i$, we can get:

$$\mathcal{B}^*(x_1, x_2, \dots, x_n | m) = (n-1) \sum_{i=1}^n \sum_{j=1}^{+\infty} N_{ij} (x_i - k_i)^j \quad (10)$$

$$= (n-1) \sum_{i=1}^n N_{i1} (x_i - k_i) + \mathcal{O}(x_i) \quad (11)$$

$$\approx (n-1) \sum_{i=1}^n N_{i1} (x_i - k_i), \quad (12)$$

³The basic form $\mathcal{B}(t_i | m)$ and the combined form $\mathcal{B}(+\infty, \dots, +\infty, t_i, +\infty, \dots, +\infty | m)$ are not completely equivalent.

where $N_{i1} = \frac{\partial \mathcal{B}^*(x_1, x_2, \dots, x_n | m)}{\partial x_i}$. Then the original formula is expressed as:

$$\mathcal{B}(t_1, t_2, \dots, t_n | m) \approx \frac{1}{(n-1) \sum_{i=1}^n \frac{N_{i1}}{t_i - b_i} - k_i} \quad (13)$$

Given the minimal change in the derivative within the observable range, N_{i1} is treated as a constant N_i in this task for simplicity. Experimental results show that, if sub-RBs are separated independently, k_i is typically 0. Since t_i cannot be directly quantified, we use basic form of $\mathcal{B}(t_i | m)$ as its quantized substitute, thus simplifying the combination law as:

$$\mathcal{B}(t_1, t_2, \dots, t_n | m) \approx \frac{1}{(n-1) \sum_{i=1}^n \frac{N_i}{\mathcal{B}(t_i | m) - b_i}} \quad (14)$$

A.3 Calculation of RB in Practical Process

To determine the constants, we first fit parameters to a model using a development dataset (or 20% of the test dataset if the development dataset is not available). This fitting process yields the corresponding constants. For a given task and prompt strategy, these constants remain fixed. Additionally, once the combination law constants are established, different reasoning boundaries are determined through a binary search on performance in a standard setting (3-shot CoT). For instance, we use binary search to identify a reasoning boundary that ensures the accuracy of all problems below that boundary approaches 90%, achieving $\mathcal{B}_{Acc=90\%}$. For one model, one task, and one prompt type, the reasoning boundary remains fixed. Zero-shot and few-shot settings share the same set of reasoning boundaries.

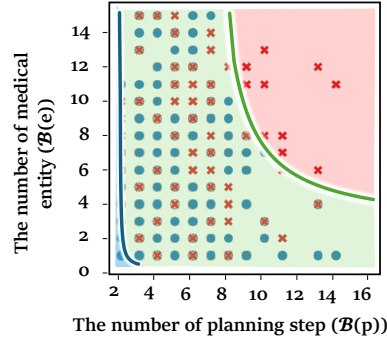


Figure 12: Extended verification of combination law on Medical Knowledge Probing [Cheng et al., 2024] tasks.

B The Application Tutorial of Reasoning Boundary

From a practical standpoint, our mechanism framework exhibits universal adaptability, making it suitable for application in a wide range of scenarios. When confronted with a new problem context, the framework enables a systematic approach to problem-solving. A key feature of the framework is its reliance on the weighted harmonic mean, which imparts advantageous mathematical properties to its structure. Specifically, the framework operates effectively if the reasoning process can be segmented into relatively independent boundaries. This segmentation allows the framework to be fully leveraged in addressing diverse problems.

Reasoning Boundary Application. In the case of a vertical domain problem based on CoT reasoning, the process can be divided into two key boundary levels: task planning and domain-specific reasoning. These can be modeled as follows:

$$\mathcal{B} = \frac{1}{\frac{1}{\mathcal{B}_p} + \frac{1}{\mathcal{B}_v} + k_1}, \quad (15)$$

where: $\mathcal{B}_p$ represents the task planning boundary, $\mathcal{B}_v$ represents the vertical domain boundary, and k_1 is a constant reflecting the degree of boundary independence.

Reasoning Boundary Definition & Segmentation. Neglecting any of these boundaries only results in an increase in k , but keeping the overall efficiency of the framework. If the reasoning boundary is well-defined and independent, the value of k approaches zero, showcasing the effectiveness of our mechanism framework.

Further Reasoning Boundary Segmentation. Further refinement of the vertical domain boundary, $\mathcal{B}_v$, into $\mathcal{B}_{v1}$ and $\mathcal{B}_{v2}$ is straightforward. No additional complexity is introduced, as the following relationship holds:

$$\mathcal{B}_v = \frac{1}{\frac{1}{\mathcal{B}_{v1}} + \frac{1}{\mathcal{B}_{v2}} + k_2}. \quad (16)$$

Thus, the overall boundary equation can be extended to:

$$\mathcal{B} = \frac{1}{\frac{1}{\mathcal{B}_p} + \frac{1}{\mathcal{B}_{v1}} + \frac{1}{\mathcal{B}_{v2}} + k_1 + k_2}. \quad (17)$$

This formulation allows for flexible and systematic boundary division at multiple levels, enhancing the framework’s practical utility across various problem domains.

Challenging Reasoning Boundary Measurement. In addition, we propose an alternative method to measure the reasoning boundaries. This approach allows the model to provide direct answers without relying on CoT reasoning steps. By doing so, the model’s reasoning process for a specific task depends solely on a single reasoning boundary, which can be represented as follows:

$$\mathcal{B} = \frac{1}{\frac{1}{\mathcal{B}_1} + k_1}. \quad (18)$$

For instance, in the MGSM task, assessing multilingual reasoning boundary is particularly challenging. To address this, we directly evaluate the model’s performance using a direct prompting strategy without CoT outputs and use this performance to define the multilingual reasoning boundary, which in turn helps determine the corresponding normalization constant. Subsequently, we apply multilingual CoT reasoning to the MGSM task to calculate the combined boundary using the framework’s combination law. This approach provides a more generalized solution and may be more adaptable to specific needs.

C Details of Dataset

C.1 Dataset Construction

To adequately assess the reasoning boundary of LLMs, it is essential to develop a dataset that encompasses a range of complexities and reasoning boundaries. To address these challenges, we propose a novel approach to constructing a mathematical reasoning dataset using manual synthesis and annotation which finally leads to the BIGGSM benchmark. Specifically, our proposed method involves the manual synthesis and annotation of a mathematical reasoning dataset. The construction process includes the following steps:

Step 1: Domain Template Generation Initially, we employ a prompt-driven LLM (GPT-4) to generate complex scenarios necessitating multi-step calculations. This process also yields initial example templates. Specifically, the prompt given to the large model is as follows:

Generate a scenario-related template involving multiple mathematical steps to solve a real-world problem. Ensure the scenario requires the application of different mathematical concepts. Please use "[VAR]" as a variable to mark the template of the question.

Step 2: Natural Language Template Creation Recognizing that LLMs can produce errors and logical inconsistencies, we refine these initial templates to improve their accuracy and add mathematical calculations. To facilitate the generation of extended sequences, we decompose the templates into smaller, loopable segments that incrementally meet the multi-step reasoning demands.

Step 3: Domain Template Augmentation To address the limited diversity in individual samples and provide a broader evaluation of LLMs’ mathematical abilities, we use an LLM (GPT-4) to generate at least three alternative augmented templates for each original template and step. The generation prompt we use is as follows:

Create three alternative versions of the following template that introduce different complexities or variables, ensuring each version demands an equivalent level of reasoning.

Step 4: Numeric Filling Once all templates are prepared, we aim to test the upper-bound of the LLMs’ computational reasoning boundary by introducing numerical values ranging from 1 to 1e5 in multiplication tasks. This step is designed to thoroughly assess the models’ performance across a spectrum of numerical challenges.

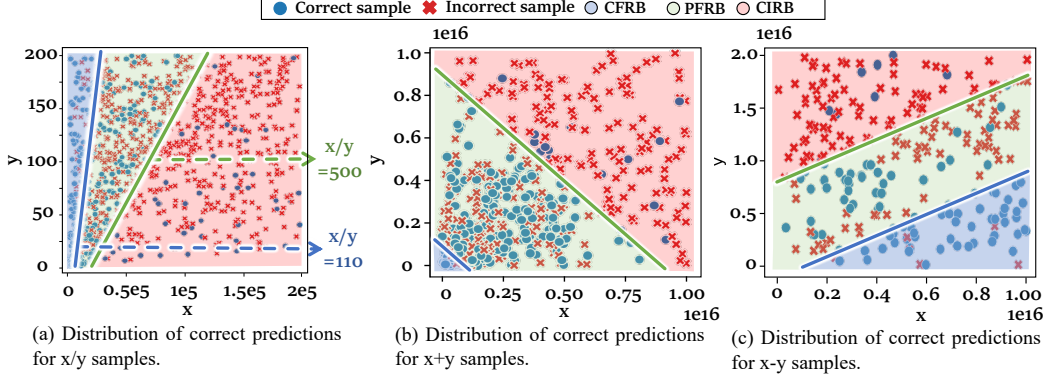


Figure 13: Existence verification for reasoning boundaries on basic arithmetic calculation tasks, including division, addition, and subtraction operations.

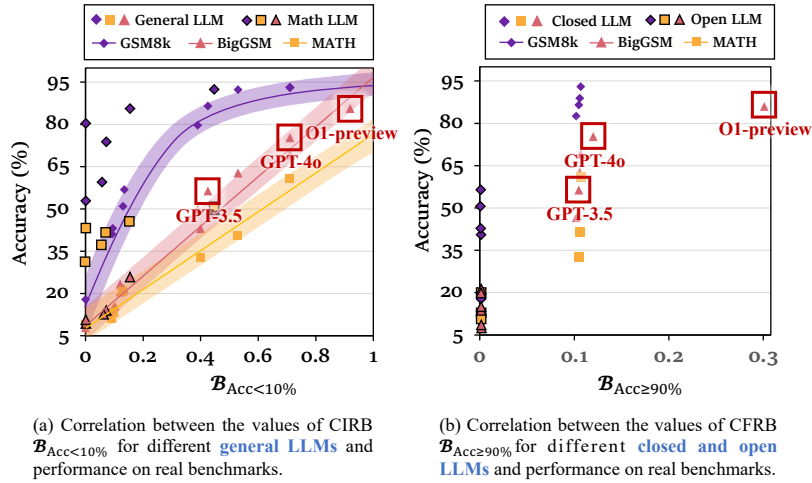


Figure 14: Correlation between the values of RB for different models and performance on real benchmarks.

Step 5: Manual Annotation To ensure the quality and logical coherence of our synthetic samples, we manually review them to correct any errors introduced during the automated generation process. Finally, we hired three experts to mark whether the samples in the data set were correct. Only for those samples where more than two experts agreed did we retain the corresponding samples. The Cohen’s kappa value marked by the experts was 0.97, which indicates the perfect agreement.

C.2 Dataset Analysis

Our dataset comprises 610 test samples, which is extensive when compared to the GSM8K dataset. It features a broader range of procedural steps, varying from 1 to 16 steps. Additionally, our dataset encompasses a wider spectrum of computational efforts, ranging from 6 to $3e5$.

D The Implementation Details of Basic Arithmetic Calculation

D.1 Data Construction

This section outlines the process used to construct datasets for examining the existence of reasoning boundaries (RB) in basic arithmetic calculations. Initially, we identify the operations for investigation, namely addition, subtraction, multiplication, and division. We then determine the range of integer operands (x and y), starting from 1 to $1e10$, subsequently extending to $1e20$. A random number generator is employed to produce independent and unbiased pairs of x and y within the specified range. For each pair, we compute the expected correct outcome of the chosen operation using standard

arithmetic procedures. In addition, in order to ensure that decimals do not affect the computational complexity, we restrict our analysis to integer operands and outcomes to control for complexity and randomly generate numerical values of x and y .

D.2 Prompt Construction

The prompt configuration in our study involves inputting the structured data into a computational model to analyze the arithmetic accuracy. The following prompting is used for LLMs' input:

Please calculate the formula given below:
 $x \text{ op } y =$

where **op** denotes the arithmetic operation (selected from addition, subtraction, multiplication, division). And x and y values are generated from Section D.1. The final experimental results are shown in Figure 13.

E The Implementation Details of Multi-hop Reasoning

We propose that the natural language multi-hop CoT task comprises two sub-tasks: multi-hop planning and knowledge step reasoning for multi-hop question answering. To address the challenge of measuring knowledge difficulty, we utilize a NER model⁴ to identify the number of knowledge entities in each hop, thus marking the knowledge step reasoning RB in the single-step task. Formally, let $\mathcal{B}(h)$ represent the RB of multi-hop planning and $\mathcal{B}(e)$ denote the RB of knowledge step reasoning. The combined RB satisfies the following combination law:

$$\mathcal{B}^{\text{CoT}}(e, h) = \frac{1}{\frac{N_1}{(\mathcal{B}(e)-b_1)} + \frac{N_2}{(\mathcal{B}(h)-b_2)}}. \quad (19)$$

F Analysis for Complex-CoT and Least-to-Most within Reasoning Path Optimization Perspective

Complex CoT Prompting can achieve better CoT within a specific RB by simplifying the calculation reasoning step. We believe that Complex CoT optimizes the performance of the model by allowing the model to reach its computational limit as much as possible in single-step reasoning. Therefore, the combined RB for Complex-CoT can be expressed as:

$$\mathcal{B}^{\text{Complex}}(p, c) = \lim_{\mathcal{B}(c) \rightarrow \mathcal{B}_{\text{Acc}=100\%}(c)} \frac{1}{\frac{N_1}{(\mathcal{B}(c)-b_1)} + \frac{N_2}{(\mathcal{B}'(p)-b_2)}} \quad (20)$$

Assuming the premises of RB remain unchanged ($\mathcal{B}^{\text{Complex}}(p, c) = \mathcal{B}^{\text{CoT}}(p, c)$), it can obviously yield the solution $\mathcal{B}'(p) > \mathcal{B}(p)$. Therefore, the model can accept more steps of reasoning boundary, that is, if the planning difficulty d_p is less than reasoning capability $\mathcal{B}'(p)$, the accuracy is higher. In order to analyze this problem, we adopted a meta-analysis method. We count the performance of the work of Jin et al. [2024], Fu et al. [2023] using Complex CoT. The relationship between the performance label and the number of steps is shown in Figure 6 (left). For most multi-step reasoning tasks, generally speaking, within a certain range, as the number of steps increases, the computational pressure of the model is relieved and the performance is improved, which is consistent with the theory and exploration of Feng et al. [2024], Wang et al. [2023b], Valmeekam et al. [2023].

However, We can also clearly recognize the flaws of Complex CoT. Once the difficulty of planning d_p (that is, the number of planning steps) is greater than $\mathcal{B}'(p)$, it exceeds the capabilities of the model and the performance will decline. We can observe that for single-step calculation reasoning, as shown in Figure 6 (right), the performance of using Complex CoT will gradually decrease. The rest of mathematical reasoning will also decrease when the number of steps is greater than a certain threshold. This phenomenon can also be explained by our combination law. While the amount of calculation is reduced, the number of reasoning steps is also increasing. If the acceptable number of reasoning steps is exceeded, the reasoning boundary is exceeded, and the model performance will

⁴<https://huggingface.co/dslim/bert-base-NER>

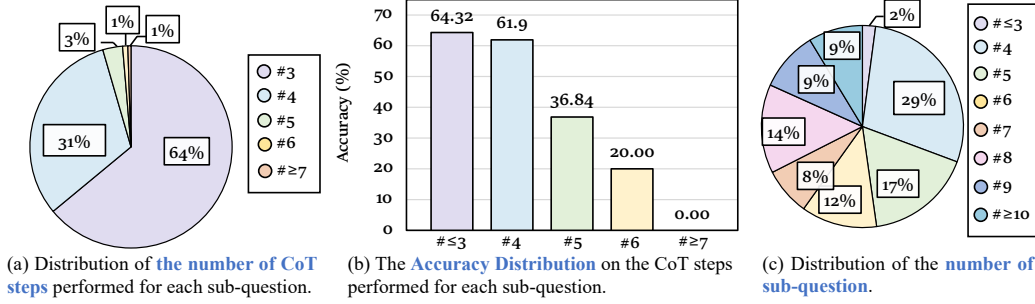


Figure 15: Analysis of output results for Least-to-Most Prompting.

decline, which demonstrates that it is necessary to keep a balance between the number of reasoning steps and computational pressure (see Appendix G for detailed meta-analysis process).

Limitation: Need to keep balance in the number of reasoning steps and calculation pressure.

Least-to-Most Prompting can achieve better CoT within a specific RB by simplifying the planning reasoning paths. Least-to-most prompting structures problem-solving hierarchically, by breaking questions into smaller sub-questions and further solving them one-by-one. Accordingly, the Least-to-most RB can be divided into three sub-RBs, namely, the problem decomposition RB $\mathcal{B}(d)$, the problem planning RB $\mathcal{B}(p)$, and the single-step calculation RB $\mathcal{B}(c)$. Therefore, the combined RB for least-to-most can be expressed as:

$$\mathcal{B}^{\text{LtM}}(d, p, c) = \frac{1}{\frac{2N_1}{(\mathcal{B}'(c)-b_1)} + \frac{2N_2}{(\mathcal{B}(p)-b_2)} + \frac{2N_3}{(\mathcal{B}(d)-b_3)}}. \quad (21)$$

Ideally, if the problem decomposition ability of the model is excellent ($\mathcal{B}(d) \rightarrow +\infty$), it can decompose the problem into sub-problems that can be solved in one step every time $\mathcal{B}(c) \rightarrow 1$, therefore the least-to-most RB can be expressed as:

$$\hat{\mathcal{B}}^{\text{LtM}}(d, p, c) = \lim_{\mathcal{B}(c) \rightarrow 1, \mathcal{B}(d) \rightarrow +\infty} \mathcal{B}^{\text{LtM}}(d, p, c) = \frac{\mathcal{B}'(c) - b_2}{N_1(\mathcal{B}'(c) - b_2) - N_2}, \quad (22)$$

Assuming the premises of RB remain unchanged ($\hat{\mathcal{B}}^{\text{LtM}}(d, p, c) = \mathcal{B}^{\text{CoT}}(p, c)$), it can obviously yield the solution $\mathcal{B}'(c) > \mathcal{B}(c)$. On the contrary, the model can accept larger difficulty d , which also shows that using least-to-most prompting can effectively increase the maximum of acceptable calculation RB under a given RB (as shown in Figure 7), thereby improving model performance. As shown in Table 1, we find that LLM can be optimized by Least-to-most from vanilla CoT.

However, the performance improvement of the model is not significant, which we attribute to the fact that the current model cannot push its performance to the ideal limit. As shown in Figure 15 (a), the reasoning boundary of the model cannot make each reasoning step completely tend to 1, which also leads to the difference in reasoning performance in Figure 15 (b). In the meantime, the model's ability to divide problems is also limited. What's more, as shown in Figure 15 (c), in around 90% of cases, the model will only divide less than 6 problems, which also limits the performance.

Limitation: Although the pressure of local planning has been reduced, it has not actually effectively reduced the pressure of global planning, nor the pressure of optimization calculations.

G The Meta-Analysis for Complex-CoT Prompting

In order to discuss the advantages and limitations of Complex-CoT, we conduct two detailed meta-analyses through two distinct perspectives. The first one assesses the influence of the reasoning steps demonstrated in In-Context Learning (ICL) across various tasks, while the second evaluates the effects of employing a fixed number of reasoning steps in ICL on questions with different reasoning steps. These meta-analyses aim to compare the efficacy of these methods against prior studies

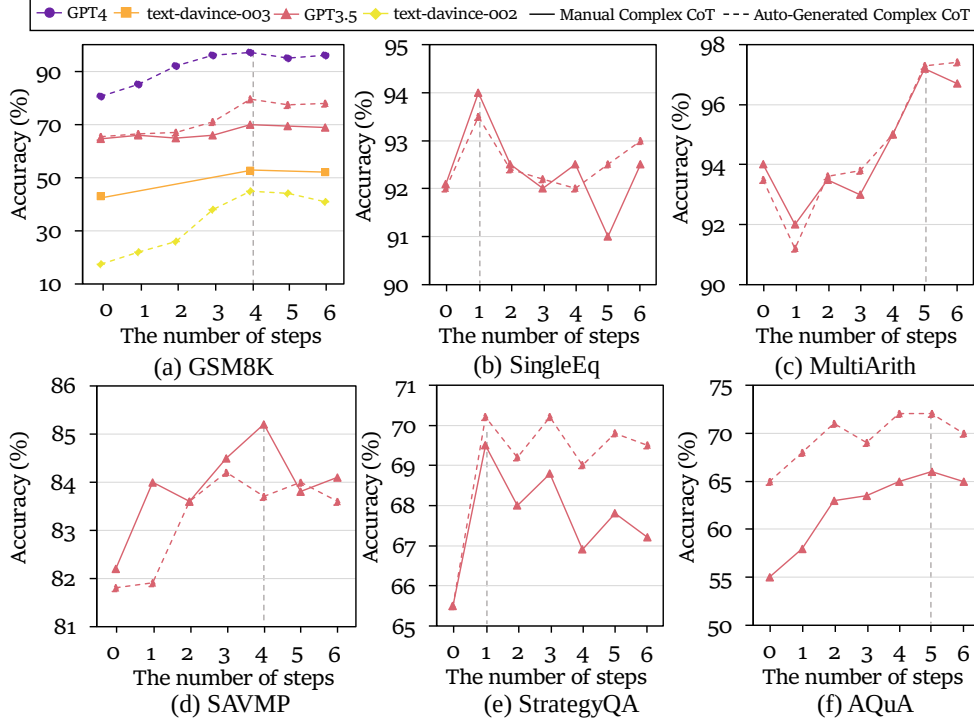


Figure 16: The effectiveness on average step length in demonstrations for CoT performance.

systematically. Specifically, we conducted a systematic search for relevant studies addressing the same problem tackled by Jin et al. [2024], Fu et al. [2023], Shum et al. [2023], Sun et al. [2023], and Jiang et al. [2023b]. We ensured that retrieved studies were pertinent, focusing on studies that addressed the same problem and used similar evaluation metrics.

G.1 The effectiveness of step length in demonstrations

From each selected study, including Jin et al. [2024] and Fu et al. [2023], we evaluate the performance using Complex CoT. The relationship between performance and the number of Complex CoT’s steps is shown in Figure 16. For most multi-step reasoning tasks, as the number of steps increases within a certain range, the computational load decreases, and performance improves.

However, as described in Appendix F, the flaws of Complex CoT are apparent. When the difficulty of planning (d_p), defined as the number of planning steps, exceeds $B'(p)$, the model’s capabilities are surpassed, leading to a performance decline. This is evident in single-step calculation reasoning, as shown in Figure 16 (c, e), where performance using Complex CoT gradually decreases. Similarly, for other mathematical reasoning tasks, performance decreases when the number of steps exceeds a certain threshold. This phenomenon aligns with our combination law: while reducing the amount of calculation, the number of reasoning steps increases. Exceeding the acceptable number of reasoning steps surpasses the reasoning boundary, causing a decline in model performance. Therefore, maintaining a balance between the number of reasoning steps and computational pressure is crucial.

G.2 The effectiveness on step length in golden samples

Furthermore, to gain a nuanced understanding of the impact of Complex CoT when the number of steps exceeds the golden step number, we conduct further meta-analysis from Fu et al. [2023], Shum et al. [2023], Sun et al. [2023], Jiang et al. [2023b]. Specifically, as illustrated in Figure 17 (a, b), our analysis reveals that for problems of low complexity and with smaller golden step numbers, Complex CoT tends to underperform compared to Vanilla CoT. Notably, it is only when the reasoning steps exceed two that Complex CoT outperforms Vanilla CoT. This suggests that Complex CoT effectively optimizes single-step computations and enhances model performance for complex

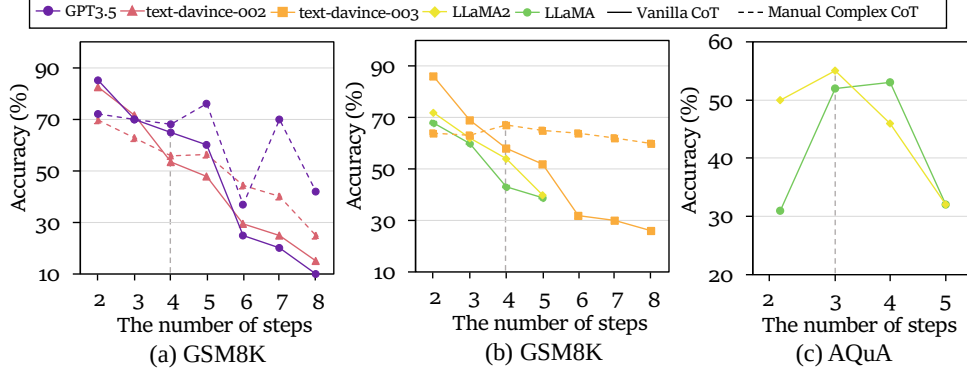


Figure 17: The effectiveness of step length in golden samples with a fixed step length in demonstrations for CoT performance.

[Minimum Reasoning Path Prompting]

You need to perform multi-step reasoning, with each step carrying out as many basic operations as possible.

[Acceptable Reasoning Prompting]

Remember, you can only complete tasks that contain up to 5 basic operations per step, and multiplication operations must be less than $1.5e5$. The upper limit of the multiplication operations decreases as the number of operations per step increases.

[EXAMPLE]

Question: Leo's assignment was divided into three parts. He finished the first part of his assignment in 25 minutes. It took him twice as long to finish the second part. If he was able to finish his assignment in 2 hours, how many minutes did Leo finish the third part of the assignment?

Answer: Leo finished the first and second parts of the assignment in $25 + 25 \times 2 = 75$ minutes.

Therefore, it took Leo $60 \times 2 - 75 = 45$ minutes to finish the third part of the assignment.

45

...

[REQUEST]

Question: [Question]

Figure 18: Minimum acceptable reasoning path prompting for natural language chain-of-thought. All examples given in the context transform from Wei et al. [2022].

problems. However, it increases the cognitive load for simple problems, resulting in a performance decline.

Interestingly, this phenomenon is also observed with the simpler Vanilla CoT, as shown in Figure 17 (c). The model achieves significant performance gains only when the number of reasoning steps aligns with the target output steps. If the complexity of the planned steps exceeds the necessary reasoning boundary, or if there is no effective optimization for reasoning boundary, the performance deteriorates.

G.3 The Implementation Details of Minimum Acceptable Reasoning Paths

To address the two aforementioned limitations, we propose Minimum Acceptable Reasoning Paths (MARPs). Firstly, to reduce the model's computational load, we introduce instructions that limit its single-step computing power, thereby optimizing its reasoning boundary. Secondly, to enhance the

Model	Acc. ($\uparrow$)	Input Token ($\downarrow$)	Output Token ($\downarrow$)
HotpotQA [Yang et al., 2018]			
CoT	289.50	67.27	26.50
CoT+MARP	309.51	68.39	28.73
Medical Probing [Cheng et al., 2024]			
CoT	636.11	249.78	48.9
CoT-MRP	476.11	86.52	69.41
StrategyQA [Geva et al., 2021]			
CoT	1046.28	225.35	63.90
CoT+MARP	649.28	167.40	74.09

Table 2: Extended experimental results on GPT-3.5-Turbo.

model’s acceptability, we increase the computation amount per step within this boundary and reduce the number of global planning steps, thus alleviating planning pressure.

To control variables effectively, we make only the simplest modifications to the prompt to achieve the desired CoT optimization.

Minimum Reasoning Path Prompting To alleviate the cognitive load associated with planning, it is essential to have the model respond to the question as succinctly as possible. This approach ensures that the focus remains on providing a short, clear and direct reasoning path. The following prompt is designed to achieve this objective:

You need to perform multi-step reasoning, with each step carrying out as many basic operations as possible.

Acceptable Reasoning Prompting To effectively utilize the model, it is crucial to define the upper-bound of reasoning boundary. This ensures that the complexity of the reasoning process is manageable and within acceptable bounds. The specific prompt to achieve this is as follows:

Remember, you can only complete tasks that contain up to 5 basic operations per step, and multiplication operations must be less than $1.5e5$. The upper limit of the multiplication operations decreases as the number of operations per step increases.

This prompt is designed to set clear boundaries for the model’s operations, thereby optimizing its performance and accuracy.

Furthermore, it is necessary to enhance the demonstration within the corresponding in-context learning framework to meet the specific needs of our Model-Agnostic Reasoning Protocol (MARP). This involves refining the examples and instructions provided to ensure they align perfectly with the MARP requirements. Figures 18 and Figure 19 illustrate our MARP prompt, showcasing how to structure the demonstrations to facilitate effective learning and reasoning in natural language CoT and program-of-thought setting. By adhering to these guidelines, we can ensure that the model operates efficiently and produces reliable results.

In summary, setting precise boundaries for reasoning boundary and optimizing in-context learning demonstrations are essential steps in enhancing the model’s performance. By following the specified prompt and refining the MARP examples, we can achieve a high level of accuracy and efficiency in the model’s reasoning processes.

H The Implementation Details in various LLMs

We employ 25 commonly used models to evaluate the extensibility of our framework to a broader range of models. The specific models are listed in Table 3. For each model, we utilize the chat/instruct version whenever available to maximize their ability to follow instructions. Additionally, we deploy all models on the vLLM [Kwon et al., 2023] framework to ensure a fair comparison. Except for model OpenMath-series [Toshniwal et al., 2024] which does not conform to the vLLM format, all other models are deployed on vLLM for testing. All experiments on open-source models were

[Minimum Reasoning Path Prompting]

You need to perform multi-step reasoning, with each step carrying out as many basic operations as possible.

[Acceptable Reasoning Prompting]

Remember, you can only complete tasks that contain up to 5 basic operations per step, and multiplication operations must be less than $1.5e5$. The upper limit of the multiplication operations decreases as the number of operations per step increases.

[EXAMPLE]

Question: Leo's assignment was divided into three parts. He finished the first part of his assignment in 25 minutes. It took him twice as long to finish the second part. If he was able to finish his assignment in 2 hours, how many minutes did Leo finish the third part of the assignment?

Answer: Leo finished the first and second parts of the assignment in $25 + 25 \times 2 = 75$ minutes.

Therefore, it took Leo $60 \times 2 - 75 = 45$ minutes to finish the third part of the assignment.

45

...

[REQUEST]

Question: [Question]

Figure 19: Minimum acceptable reasoning path prompting for program-of-thought. All examples given in the context transform from Wei et al. [2022].

conducted on two A100 80G. Following the setting of Wei et al. [2022], in our CoT experiment, all multi-step reasoning tasks are with three manually constructed demonstrations. In addition, for all the experiments, our top-p is selected from $\{0.95, 1\}$, and temperature is selected from $[0, 1]$.

In addition, the only difference in the prompt is that we use different dialogue delimiters to make it conform to the format of the LLM instruction fine-tuning, thereby avoiding the bias caused by the gap between training and inference.

Model	Base Model	Parameters (B)
<i>Open-source General LLM</i>		
LLaMA [Touvron et al., 2023a]	-	7, 13, 33, 65
LLaMA-2 [Touvron et al., 2023b]	-	7, 13, 70
LLaMA-3 [Meta, 2024]	-	8, 70
Code-LLaMA [Roziere et al., 2023]	LLaMA-2 [Touvron et al., 2023b]	7, 13, 34, 70
Mistral [Jiang et al., 2023a]	-	7
<i>Close-source General LLM</i>		
Gemini-1.0-Pro [Team et al., 2023]	-	-
GPT3.5-Turbo [OpenAI, 2022]	-	-
Claude-3-Haiku [Anthropic, 2024]	-	-
Claude-3-Sonnet [Anthropic, 2024]	-	-
Claude-3-Opus [Anthropic, 2024]	-	-
GPT4 [OpenAI, 2023]	-	-
<i>Open-source Math LLM</i>		
MAMmoTH [Yue et al., 2023]	LLaMA-2 [Touvron et al., 2023b]	7, 13
MAMmoTH [Yue et al., 2023]	Mistral [Jiang et al., 2023a]	7
OpenMATH-Instruct [Toshniwal et al., 2024]	LLaMA-2 [Touvron et al., 2023b]	70
OpenMATH-Instruct [Toshniwal et al., 2024]	Mistral [Jiang et al., 2023a]	7

Table 3: Model list. In order to ensure a certain ability to follow instructions, we use the Instruct version of the model as much as possible (if available).

I The Implementation of Combination Law in MGSM

Inspired by [Qin et al. \[2023\]](#) and [Huang et al. \[2023\]](#), we propose that the multilingual mathematical CoT task comprises three sub-tasks: step planning, step calculation, and multi-modal expression. We evaluate the model’s mathematical expression ability in different languages based on its zero-shot direct performance on MGSM, as reported by [Qin et al. \[2023\]](#). For relevant parameter calculations, please see the “Challenging Reasoning Boundary Measurement” part of Appendix B. Formally, let step planning RB be denoted by $\mathcal{B}(p)$, step calculation RB by $\mathcal{B}(c)$, and multilingual expression RB by $\mathcal{B}(l)$. The combined RB satisfies the following law:

$$\mathcal{B}^{\text{CoT}}(c, p, l) = \frac{1}{\frac{2N_1}{(\mathcal{B}(c)-b_1)} + \frac{2N_2}{(\mathcal{B}(p)-b_2)} + \frac{2N_3}{(\mathcal{B}(l)-b_3)}}. \quad (23)$$

As shown in Figure 10, the performance distribution of RB (including $\mathcal{B}_{Acc=90\%}$ and $\mathcal{B}_{Acc=10\%}$) in the multilingual mathematical reasoning task aligns with the proposed combination law in Equation (23). Moreover, three distinct RBs are evident in Figure 10.

J Ethical Considerations

Data Access. Our data is adapted from GSM8K [[Cobbe et al., 2021](#)] and supplemented with manually created samples. GSM8K is an open-source dataset available for academic research.

Dataset Collection Process. We began with an introductory task interview using 50 example questions, compensating participants \$20 each to familiarize themselves with the task. During the annotation process, annotators were paid \$15 per hour, totaling approximately 60 hours of work.

The Rest of Data Annotation Process. For the remaining data annotation, we hired a graduate student with CET-6 proficiency in Chinese and English and strong mathematical knowledge. The student was compensated \$15 per hour, which is above the local average salary. The instructions for annotation are as follows:

You need to annotate the generated number of steps, maximum computation amount, correctness of the generation steps, correctness of the calculations, and correctness of the model output:

- **Number of generated steps:** This refers to how many reasoning steps the model generated.
- **Maximum computation amount:** This indicates the largest product of operations in the model’s reasoning steps.
- **Correctness of generation steps:** This assesses the accuracy of the model’s planning. If all steps and operators are planned correctly, and the operand values are logically correct, it is considered correct, regardless of calculation accuracy.
- **Correctness of calculations:** This considers only whether the calculations are correct, ignoring planning factors.
- **Correctness of the output:** This checks whether the model’s final answer is correct.

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Answer: [Yes]

Justification: As shown in Appendix C to Appendix I, we have provided detailed descriptions and analyses of the experimental setups for all our investigations.

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 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

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Justification: As shown in Section C to Section H, we have provided detailed descriptions and analyses of the experimental setups for all our investigations.

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Justification: We report error bars in Table 1 and Figure 4, and explain the error variables in Section 3. However, error bars are not reported for all tasks because it would be too expensive for human annotation and computational resource consumption.

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LEARNING FROM LESS: MEASURING THE EFFECTIVENESS OF RLVR IN LOW DATA AND COMPUTE REGIMES

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Armin Parchami¹ Frederic Sala^{1,2} Paroma Varma¹

ABSTRACT

Fine-tuning Large Language Models (LLMs) typically relies on large quantities of high-quality annotated data, or questions with well-defined ground truth answers in the case of Reinforcement Learning with Verifiable Rewards (RLVR). While previous work has explored the benefits to model reasoning capabilities by scaling both data and compute used for RLVR, these results lack applicability in many real-world settings where annotated data and accessible compute may be scarce. In this work, we present a comprehensive empirical study of open-source Small Language Model (SLM) performance after RLVR in low data regimes. Across three novel datasets covering number counting problems, graph reasoning, and spatial reasoning, we characterize how model performance scales with dataset size, diversity, and complexity. We demonstrate that (1) procedural datasets allow for fine-grained evaluation and training dataset development with controllable properties (size, diversity, and complexity), (2) under RLVR, models trained on lower complexity tasks can generalize to higher complexity tasks, and (3) training on mixed complexity datasets is associated with the greatest benefits in low data regimes, providing up to $5\times$ sample efficiency versus training on easy tasks. These findings inspire future work on the development of data scaling laws for RLVR and the use of procedural data generators to further understand effective data development for efficient LLM fine-tuning.

1 INTRODUCTION

Recent advances in Large Language Models (LLMs) have achieved significant improvements in reasoning capabilities (OpenAI, 2025a; Comanici et al., 2025; OpenAI, 2025b; Zeng et al., 2025; Anthropic, 2025); this has, in part, been driven by the adoption of Reinforcement Learning with Verifiable Rewards (RLVR) (Shao et al., 2024; Wen et al., 2025). RLVR provides an effective method for post-training LLMs by rewarding models based on verifiable outcomes (e.g., answers that can be compared to a known unambiguously correct ground truth) rather than noisy human preferences (Poddar et al., 2024). For classes of problems with verifiable outcomes, such as in mathematics (Wang et al., 2025b), the adoption of RLVR has enabled models (e.g., DeepSeek R1 (Guo et al., 2025)) to achieve state-of-the-art performance and allowed strong problem solving and self-correction capabilities to emerge. However, many of these advances are made under the assumption that high-quality training data and compute are abundant (Khatri et al., 2025).

Recent RLVR research has focused on using high volumes of question-answer pairs to improve the reasoning capabilities of LLMs. For example, DeepMath-103K includes over 100,000 challenging and decontaminated samples for training (He et al., 2025). However, in realistic, resource-constrained situations, where both annotated data and compute may be limited, these results may be challenging to replicate or extend to new reasoning domains. While previous studies have explored scaling RLVR with respect to model size and compute budget (Khatri et al., 2025; Tan et al., 2025), or focused on reducing compute requirements (both through Small Language Model (SLM) fine-tuning (Dang & Ngo, 2025) and Low-Rank Adaptation (LoRA) (Wang et al., 2025a)), there has been less attention on data scaling and the effectiveness of RLVR in low data regimes. In this work, we focus on characterizing how the size, diversity, and complexity of training data influence the reasoning capabilities of models. Motivated by these ideas, we specifically investigate the research question: “How does model performance evolve when training data and compute are limited, and what characteristics of data impact generalization in such regimes?”

We conduct a systematic empirical study using open-source SLMs fine-tuned using RLVR under low data and compute regimes to help understand these relationships, the results

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of which help to shape data scaling laws that describe the effectiveness of RLVR in different scenarios. For our experiments, we introduce three novel datasets that allow data to be generated procedurally with desired volume, diversity, and complexity. These procedural datasets allow us to better isolate the influence of different attributes (e.g., topic of questions, complexity of questions, etc.) and study fine-tuning dynamics in controlled settings. We summarize our contributions and findings as follows:

- **New procedural datasets for reasoning tasks.** We develop three new datasets designed to support RLVR that cover number counting problems, graph reasoning, and spatial reasoning. For each of the datasets, we report empirical solving rates across 10 models (including open-source and proprietary LLMs), demonstrating the value of using procedural data in creating challenging and complex tasks.
- **Study of RLVR in low data regimes.** Using an open-source LLM (Qwen3-4B), we investigate the relationship between dataset size and composition (focusing on task complexity) and performance after fine-tuning. Different training data configurations are used, capturing a range of sizes and complexities across the three dataset types. We observe that models trained on small volumes of lower complexity tasks (i.e., easier questions) generalize to more complex tasks, and that training on a mixed complexity dataset is associated with up to $5\times$ the sample efficiency under the same data budget.

2 RELATED WORK

2.1 Scaling Laws for Language Models

Early work by Kaplan et al. (2020) and Hoffmann et al. (2022) established predictable relationships between model performance, size, data, and compute, showing that increasing training data and parameters yields smooth performance improvements under fixed budgets. These analyses primarily characterize how performance scales with model size and compute, but do not address the effect of dataset composition, particularly difficulty distribution, under fixed budgets. Zhang et al. (2024) extended these ideas to supervised fine-tuning, showing that downstream loss depends jointly on fine-tuning data size, model size, and parameter-efficient adaptation methods.

2.2 Reinforcement Learning Scaling for LLMs

Recent work has explored how RL affects post-training performance in LLMs. Khatri et al. (2025) present ScaleRL, a large-scale framework characterizing RL performance under different compute budgets and algorithmic choices, providing valuable insights into efficiency at scale. However, their study primarily emphasizes compute scaling rather than

data composition or limited-data regimes. Tan et al. (2025) analyze RL post-training in mathematical reasoning tasks using Qwen2.5 models (0.5B–14B), finding that larger models achieve higher sample efficiency and that moderate data reuse (≤ 25 epochs) approaches the performance of unique data. However, their analysis is confined to a single domain and does not explore dataset composition or difficulty effects. Lai et al. (2025) provide a comprehensive overview of post-training methods, including supervised fine-tuning and RL from feedback. Their taxonomy outlines general scaling trends but remains largely descriptive, underscoring the need for empirical studies focused on data composition rather than model or compute scaling.

2.3 Data Efficiency and Selection in RL

Efficient data utilization has been a recurring challenge in RL fine-tuning. Shen et al. (2025) identify two major bottlenecks in RLHF data scaling—reward hacking and reduced response diversity—and propose hybrid reward systems and prompt-selection strategies that emphasize harder, low-reward prompts to stabilize training and improve reasoning. Their focus on selecting informative examples aligns with our investigation into how dataset composition and difficulty influence RL performance under limited data. Li et al. (2025) introduce LIMR (Less is More for RL), showing that small amounts of carefully curated data can outperform larger datasets. By quantifying each sample’s contribution to learning, they demonstrate that 1.4K selected samples can match the performance of 8.5K unfiltered ones on mathematical reasoning tasks. While they provide a selection heuristic, they do not analyze how varying dataset size or difficulty affects performance, which our study examines.

2.4 Verification and Reasoning

Work on verifiable rewards has advanced understanding of how structured feedback can drive reasoning improvements. Wen et al. (2025) demonstrate that RLVR expands genuine reasoning ability rather than simply improving sampling efficiency, introducing CoT-Pass@K metrics to measure both reasoning and answer correctness. Liu et al. (2025) propose RISE, an online RLVR framework that jointly optimizes problem-solving and self-verification, improving verification accuracy and test-time robustness. These insights inform our reward design, which combines correctness verification with format and conciseness components to provide denser feedback in low-data training settings.

Our study complements these lines of research by holding model size and compute fixed and isolating the effect of data composition on RLVR effectiveness, motivating future work on budget-aware RLVR theory that captures interactions between optimization budget, token limits, and reward sparsity.

3 METHODOLOGY

3.1 Datasets

To evaluate model performance with controlled training data, we designed three programmatically generated datasets: (1) Counting Problems, (2) Graph Reasoning, and (3) Spatial Reasoning. Each dataset was constructed using pre-defined code templates that allow for parameterization and controlled variation across a well-defined taxonomy of operators, ranges, and/or conditions. All data instances contain verifiable outcome-level ground truth, enabling both quantitative evaluation and use in RLVR training pipelines without relying on costly annotation or less reliable verification methods (e.g., LLM-as-judge evaluation). The dataset taxonomies are hand-crafted to promote diversity through meaningful variation rather than to provide statistical guarantees of coverage. Across all three datasets, the procedural generators control multiple correlated instance properties simultaneously (e.g., graph size and edge density, operator families and step counts, number of actions and query types), so comparisons between Easy and Mixed training configurations reflect variation along several dimensions rather than a single axis of complexity.

3.1.1 Counting Problems Dataset

The Counting Problems dataset is a procedurally generated benchmark designed to evaluate the numerical reasoning and pattern recognition capabilities of language models in constrained computational tasks. The dataset employs a templating methodology to generate questions with deterministic programmatic ground-truth answers, enabling precise control over problem complexity and systematic evaluation of model performance. Each question is composed of:

- A natural language prompt that specifies a counting task over a sequence of integers within a defined range.
- A sequence of conditional filters and transformations to apply before the final counting operation.
- A deterministic ground-truth answer computed by programmatically executing the specified operations.

This design allows for direct assessment of multi-step reasoning and numerical manipulation, as models must accurately parse operation sequences, track intermediate states, and perform multi-step reasoning to produce correct answers.

Controllable Complexity. We control problem complexity through multiple structural dimensions. Primarily, we vary the *range scale*, which controls the magnitude of integer ranges from which values are drawn. We also manipulate the *operator diversity* through a taxonomy of counting and aggregation operations:

- **Basic Counting:** Count, Unique Count, Zero Count.

- **Conditional Counting:** Even Count, Odd Count, Positive Count, Negative Count, Divisible By N Count.
- **Threshold-Based:** Below Threshold Count, Above Threshold Count.
- **Arithmetic Aggregation:** Sum, Product, Mean, Median, Mode.
- **Extrema Operations:** Min, Max, Range.
- **Bitwise Operations:** Bitwise AND, Bitwise OR, Bitwise XOR, Bitwise NAND.

Additionally, we vary *compositional depth* through the number of conditional filters (1–4) and transformations (0–3) applied before the final operation, creating problems with 1–7 total intermediate steps. This yields a spectrum from simple counting to complex multi-step compositional reasoning.

Example. The following is an example counting problem.

Question: Consider the integers from 1 to 100, inclusive. First, keep only the numbers that are even. Then, keep only the numbers that are divisible by 3. Of these numbers, count how many values remain.

Solution: 16 (computed programmatically by executing: $|\{x \in [1, 100] : x \bmod 2 = 0 \wedge x \bmod 3 = 0\}| = 16$)

Evaluation Protocol. Each model receives one prompt per question and generates a completion containing reasoning (optional) and a final numerical answer. Numeric responses are parsed using regular expressions and programmatically validated via exact-match comparison against the deterministic ground truth.

3.1.2 Graph Reasoning Dataset

The Graph Reasoning dataset is a procedurally generated benchmark designed to evaluate the mathematical and spatial reasoning capabilities of language models over graph-structured problems. The dataset extends the templating methodology to formal graph-based domains, allowing precise control over question complexity and solution verifiability. Each question is composed of:

- A natural language operator that defines a computation over a graph (e.g., “Find the minimum vertex cover of an undirected graph”).
- A graph structure encoded textually as lists of nodes and edges.
- A verifiable ground-truth solution, computed via deterministic algorithms.

This design allows for direct assessment of multi-hop reasoning and long-context tracking, as models must parse graph representations and perform symbolic reasoning to produce correct answers.

Controllable Complexity. We control problem complexity through several factors. Primarily, we vary the *graph size*

(number of nodes and edges), ranging from small (5 nodes) to large (25 nodes) graphs, which increases the reasoning load and relational tracking required. We also vary the *operator diversity*, drawing from a predefined taxonomy of graph-theoretic operations that span multiple problem families:

- **Subgraph Optimization:** Minimum Density Subgraph, Maximum Clique, Maximum Independent Set, Minimum Vertex Cover, Maximum Induced Bipartite Subgraph, Acyclic Subgraph, Dense Subgraph Variants.
- **Graph Partitioning:** Balanced Cut.
- **Feedback Set Problems:** Feedback Vertex Set, Feedback Edge Set.
- **Path Problems:** Longest Path, Hamiltonian Path, Hamiltonian Cycle.
- **Graph Metrics:** Graph Diameter, Graph Radius, Graph Density.

Additionally, some instances include *weighted* and *directed* edges to introduce further structural variation, though these are not treated as primary experimental variables.

Example. The following is an example question.

Question: Find the maximum independent set of an undirected graph with 5 nodes. Find the largest set of vertices with no edges between them. If multiple maximum independent sets exist, return any one of them.

Graph: Nodes: [0, 1, 2, 3, 4]; Edges: [(0,2), (0,4)].

Solution: The maximum independent set is [1, 2, 3, 4].

Evaluation Protocol. Each model receives one prompt per question and is required to generate both a complete reasoning trace and a final answer. The responses are subsequently normalized by a secondary model (GPT-4o) into a canonical internal representation. A programmatic validator, implemented using graph-based libraries (e.g., `networkx`), then verifies each output for correctness against the ground truth or determines whether it constitutes a valid equivalent solution.

3.1.3 Spatial Reasoning Dataset

The Spatial Reasoning dataset evaluates spatial reasoning capabilities of language models. We generate the problems with varying difficulty level following the spatial reasoning setup introduced in (Dsouza et al., 2025). Similar to the graph reasoning and counting problems, this setting extends the templating methodology and allows precise control over question complexity and solution verifiability. Each question is composed of:

- A description of a 2D spatial reasoning environment consisting of a square grid (board) and a set of particles on the board. The board and particles are located in a 2D space and are oriented towards one of the cardinal directions

(East, North, West, South).

- A sequence of movement and rotation actions applied to the board and particles.
- A query about the absolute or relative location or orientation of the entities (board or particles) after all actions have been applied.
- A verifiable ground-truth solution, obtained by programmatically executing the simulation.

These problems test LLMs’ abilities to track and reason over the location and orientation of entities in 2D space. The division of problems between absolute and relative is motivated by the fundamental dichotomy of egocentric (relative) and allocentric (absolute) spatial reasoning depending on the frame of reference (Denis, 2017).

Controllable Complexity. We control the complexity via the number of actions and the type of query. Intuitively, the problems with more actions and the ones based on the relative spatial reasoning are expected to be more complex.

Example. The following is a sample problem.

Question: Consider a square grid of size 20×20 centered at (0, 0). It has two particles P1 and P2 at locations $(-1.5, 2.5)$ and $(3.5, 1.5)$, respectively. P1 and P2 face towards East and West, respectively. P1 moves 1 step forward and P2 moves 1 step backwards. What is the location of P1, relative to P2?

Solution: The location of P1 relative to P2 is $(-5.0, 1.0)$.

Evaluation Protocol. Models are prompted with the question and asked to generate a completion containing an optional explanation and answer in structured (JSON) format. If the original response has parsing errors, we fall back to parsing with a secondary model (GPT-4o). The structured response is then compared against the ground truth. For floating-point numbers, we match to the first 3 decimal places and perform an exact match for integer and string values.

3.2 Curation

For our model training experiments, we generated over 1,500 programmatically defined problems for each dataset described above. Following dataset generation, we conducted model-based evaluation runs based on each dataset’s evaluation protocol across 10 diverse foundation models spanning several model families (GPT, Claude, Gemini, Grok, Llama, and Qwen).

Each model was evaluated with a single inference call per data point, and aggregate pass rates were computed at the instance level. Figure 1 shows each dataset’s performance across the ten LLMs we used. To enable finer-grained difficulty control, we categorized all problems into three difficulty tiers based on the percentage of models that answered

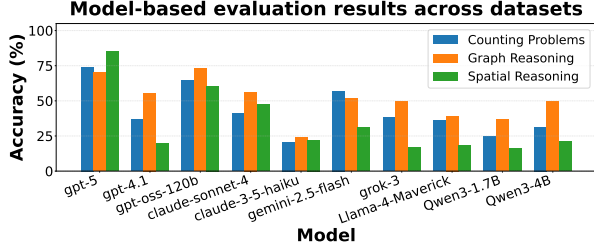


Figure 1. Overall model-based evaluation results across all generated samples for Counting Problems, Graph Reasoning, and Spatial Reasoning.

correctly:

- **Easy:** 67–100% of models answered correctly.
- **Medium:** 34–66% of models answered correctly.
- **Hard:** 0–33% of models answered correctly.

We then curated multiple subsets for downstream use by sampling based on difficulty. The following dataset configurations were curated:

- **Easy training subsets:** 100, 200, and 500 examples sampled from the Easy tier.
- **Mixed training subsets:** 100, 200, and 500 examples drawn across Easy, Medium, and Hard tiers (approximately 33% each).
- **Test subset:** 200 examples (500 examples for Graph Reasoning), not used in training, drawn across Easy, Medium, and Hard tiers.

This multi-model calibration and stratified curation distinguishes difficulty based on performance across a diverse range of models, ensuring that difficulty labels reflect actual model capabilities across architectures rather than human assumptions. Because multiple instance properties covary with difficulty tier, observed performance differences between Easy and Mixed configurations should not be attributed to a single factor. The result is a set of practical datasets that capture realistic capability boundaries across the current LLM frontier, enabling controlled studies of how problem difficulty interacts with fine-tuning dataset scale. All training, validation, and test splits are strictly disjoint to prevent data contamination or leakage between stages.

3.3 Fine-tuning

We employ RLVR to fine-tune SLMs on each problem domain. This approach enables direct optimization of task-specific reward signals rather than relying on supervised demonstrations, allowing the model to learn through exploration and self-correction.

Base Model and Architecture. We use Qwen3-4B (Yang et al., 2025) as our base model across all experiments, a

4-billion parameter model with strong reasoning capabilities. To enable efficient fine-tuning on consumer hardware, we apply LoRA (Hu et al., 2022) with rank $r = 64$ and $\alpha = 16$, targeting all linear layers. Recent work shows LoRA matches full fine-tuning performance for reinforcement learning even at low ranks (Schulman & Lab, 2025), supporting its use in our compute-constrained setting. This reduces trainable parameters to $\sim 100M$ while preserving model expressiveness.

Training Algorithm. We use Group Relative Policy Optimization (GRPO) (Shao et al., 2024), a batch-wise advantage estimation algorithm designed for mathematical reasoning. For each training example, we generate diverse completions and compute advantages by comparing rewards within each group. This approach reduces variance compared to single-sample methods while maintaining exploration. We optimize using AdamW with gradient clipping (max norm 1.0) and apply a cosine learning rate schedule with 10% warmup.

Reward Functions. We design task-specific reward functions that balance correctness, reasoning quality, and response format. We tried several formulations and report the ones that performed best so far, though they are not necessarily optimal and warrant further exploration.

Counting Rewards. Multi-component reward ($r \in [-0.4, +1.1]$) combining binary correctness ($r = 1.0$ correct, $r = 0.0$ incorrect), format quality bonuses (+0.1 for “Answer: X” format, +0.05 for acceptable variants, down to -0.1 for invalid format), and reasoning step penalties (-0.1 per step beyond 5 steps, capped at -0.3). These components apply to both correct and incorrect answers, creating positive rewards for well-formatted responses and negative rewards ($r \in [-0.4, 0]$) for verbose incorrect answers.

Graph Reasoning Rewards. Structured reward ($r \in$

Table 1. Reinforcement learning hyperparameters across datasets.

Hyperparameter	Counting	Graph Reasoning	Spatial Reasoning
Shared Architecture Parameters			
Base Model	Qwen3-4B		
LoRA Rank	64		
LoRA Alpha	16		
Training Parameters			
Training Steps	300	300	1000
Learning Rate	5×10^{-5}	5×10^{-5}	5×10^{-5}
Batch Size (per GPU)	2	1	1
Num GPUs	4	4	4
Effective Batch Size	8	4	4
Generation Parameters			
Generations per Prompt (K)	8	8	5
Temperature (τ)	1.0	1.0	1.0
Max Prompt Length	4096	4096	4096
Max Completion Length	2048	2048	2048
Optimization Parameters (Shared)			
Optimizer	AdamW		
Gradient Clipping	1.0		
LR Schedule	Cosine with 10% warmup		
Evaluation Frequency	Every 50 steps		

$[-0.2, +1.1]$) combining binary correctness ($r = 1.0$ correct, $r = 0.0$ incorrect) and format quality bonuses ($+0.1$ for proper `{"answer": ...}` JSON format). Incorrect but well-formatted responses receive partial credit ($r = 0.1$), while unstructured or excessively long outputs incur penalties ($r = -0.2$). This design reuses benchmark validation for consistent correctness evaluation.

Spatial Reasoning Rewards. Binary exact-match reward ($r \in \{0, 1\}$) using query-specific validation methods. Answer extraction supports flexible JSON formatting in multiple patterns (direct objects, code blocks) to accommodate diverse model response strategies.

Training Configuration. Table 1 summarizes our training hyperparameters. All models are trained on $4 \times$ NVIDIA A100 80GB GPUs, with training times ranging from 5–12 hours depending on problem complexity and dataset size.

Evaluation Protocol. During training, we monitor validation performance every 50 steps on a held-out 10% split from the training distribution. After training is complete, we evaluate on the curated test subset (Section 3.2). We use greedy decoding (temperature 0) for test evaluation to assess the model’s most confident predictions. Test accuracy serves as our primary metric for comparing scaling behaviors across dataset sizes and difficulty distributions.

4 RESULTS

We performed a range of RL fine-tuning experiments with the curated datasets and training setup outlined in the previous section. Our goal is to empirically compare data curation strategies under fixed training constraints, rather than to isolate a single causal mechanism. In this section, we first break down the training results on a per-dataset basis, then discuss the high-level implications of fine-tuning on programmatically generated data. Aggregate test accuracies are reported in Table 2; per-difficulty breakdowns are provided alongside each dataset’s training curves.

4.1 Dataset Results

4.1.1 Counting Problems

Training-Time Observations. Figure 2a shows the training progression for both easy-difficulty and mixed-difficulty. Under a fixed 300-step training budget, models trained on mixed-difficulty data overall showed continued validation improvement through step 300, suggesting that larger datasets may require proportionally more training steps—under compute constraints, small diverse datasets outperform large homogeneous ones.

Easy-difficulty training showed greater variation across dataset sizes. The 100-example model exhibited severe instability, peaking at 0.89 validation reward (step 150) be-

fore declining to 0.59 (step 300), coinciding with gradient norm spikes exceeding $850 \times$ baseline values (Figure 4). The 200- and 500-example models trained stably (final rewards 0.90 and 0.68), suggesting a minimum dataset size threshold between 100 and 200 examples for stable optimization. Notably, Mixed-100 trained stably despite the same sample count, indicating that difficulty diversity can substitute for dataset size in stabilizing RLVR training under limited data and compute. To better understand what drives aggregate reward variation, we decompose completions into correctness and format categories (Figure 5). For counting, the breakdown confirms that the Easy-100 collapse is driven by a drop in correctness rate rather than format or extraction issues.

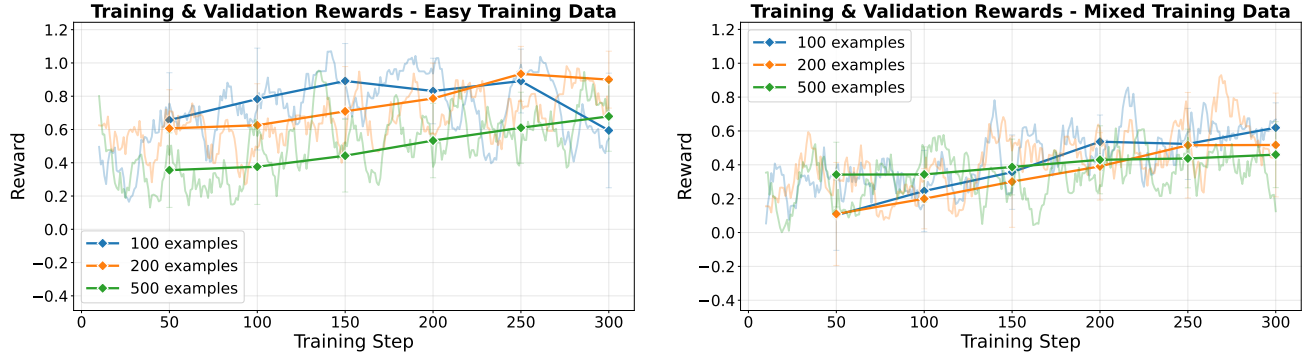
Test-Time Generalization. Test performance confirmed consistent improvement for the easy models, and showed a degradation in performance for mixed models. Mixed models achieved 0.478 (100 examples), 0.476 (200 examples), and 0.367 (500 examples) mean reward, with corresponding solving accuracies of 50.0%, 50.5%, and 40.0%. Easy-trained models scaled monotonically from 0.218 to 0.461, but required $5 \times$ more examples to match mixed baselines (500 easy examples $\approx$ 100 mixed examples in final accuracy).

Figure 3a decomposes test performance by question difficulty. Mixed-trained models show performance degradation with scaling. In contrast, easy-trained models scale monotonically. The 100-example mixed model maintains the most balanced cross-difficulty profile, while easy-trained models require 500 examples to match mixed-trained performance on easy questions.

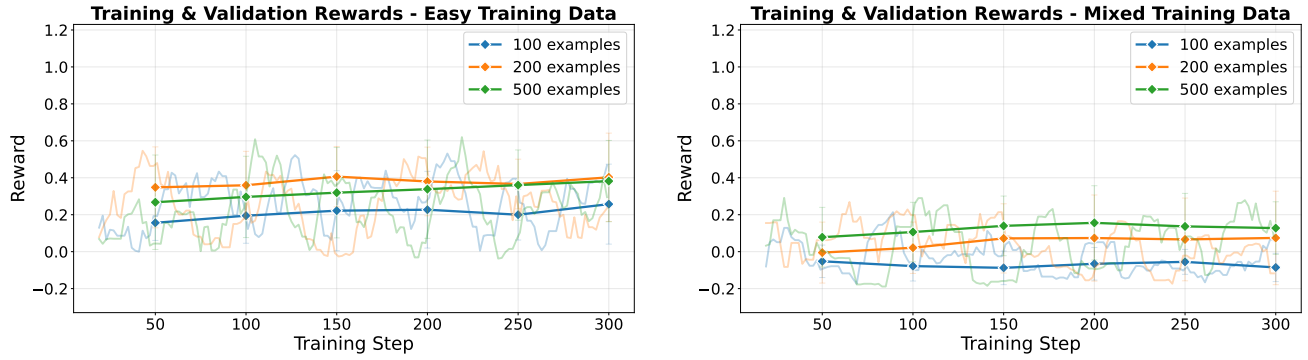
Results demonstrate two novel scaling behaviors: (1) *Degraded scaling under compute constraints*—however, with continued validation improvement toward the final step, we hypothesize that with more compute, we could see potential inverted-U scaling, contradicting supervised fine-tuning scaling laws that predict monotonic improvement (Zhang et al., 2024); and (2) *$5 \times$ sample efficiency of diverse training data*, suggesting that data composition may outweigh data quantity in low-resource RL regimes. These findings suggest that practitioners facing compute constraints should prioritize dataset diversity over size.

4.1.2 Graph Reasoning

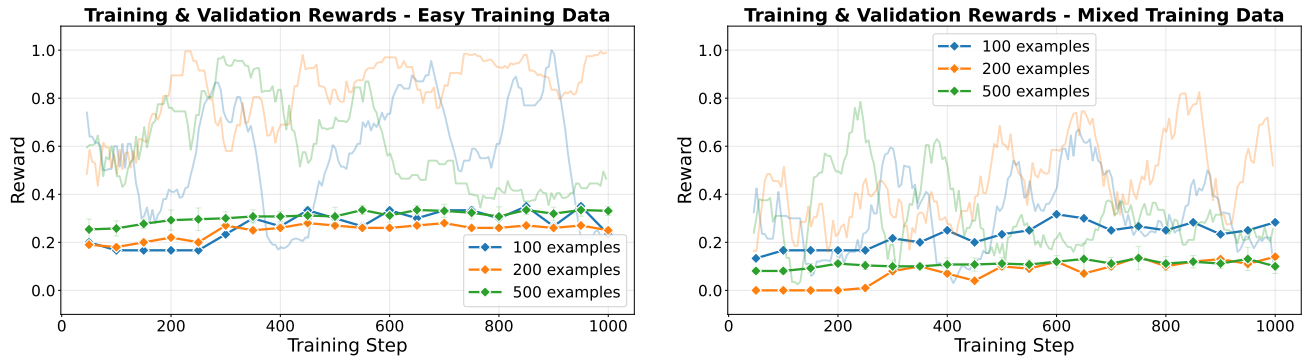
Training-Time Observations. Graph reasoning showed stable but modest improvement in validation performance across all training runs, except for the *Mixed-100* configuration (Figure 2b). In contrast, all easy-only runs showed mostly consistent upward reward trajectories through step 300, indicating continued learning despite high training reward variance. As shown in Table 2, the *Easy-500* model achieved the strongest test performance overall, suggesting



(a) Counting: training and validation reward over 300 steps. Note the instability in the easy 100-example model (collapse after step 200).



(b) Graph: training and validation reward over 300 steps. Easy-only training (left) achieves positive validation rewards; mixed-difficulty (right) shows consistently negative rewards due to incomplete rollouts under token constraints.

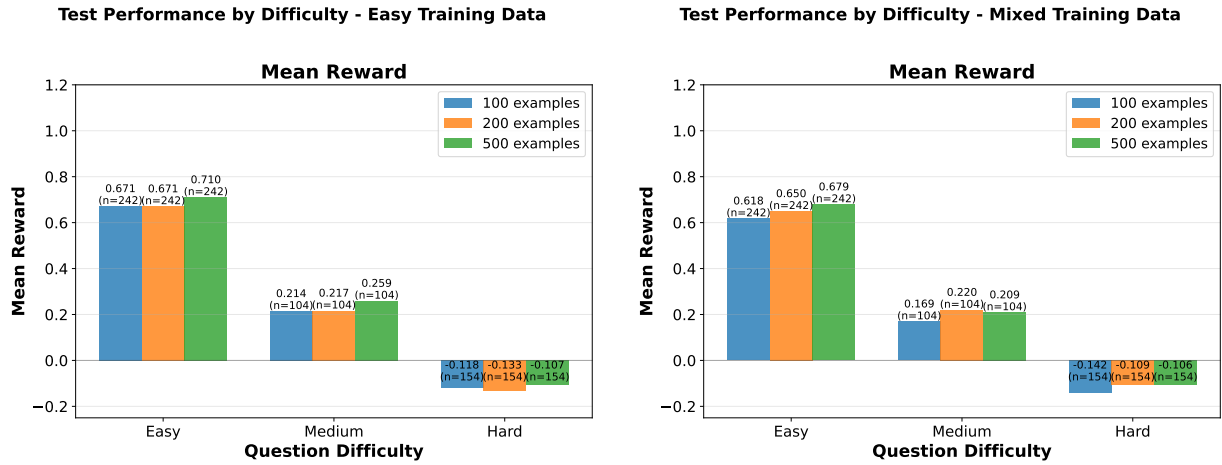


(c) Spatial: training and validation reward over 1000 steps. Binary reward ($r \in \{0, 1\}$) creates discrete performance levels. Both regimes show steady improvement.

Figure 2. Training reward curves across all three datasets (left: easy-only, right: mixed-difficulty). Colors: blue = 100, orange = 200, green = 500 examples. Light shaded lines show training rewards; dark solid lines with diamond markers show validation rewards.



(a) Counting: easy-trained models (left) specialize on easy questions but fail on harder ones; mixed-trained models (right) maintain consistent cross-difficulty performance.



(b) Graph: both training regimes achieve positive rewards on easy questions but struggle with medium and hard problems due to token constraints.



(c) Spatial: discrete binary rewards yield modest cross-distribution differences, with both models generalizing similarly.

Figure 3. Test accuracy by question difficulty across all three datasets (left: easy-trained, right: mixed-trained). Colors: blue = 100, orange = 200, green = 500 examples. Bars show accuracy on the held-out test set.

Samples	Counting		Graph		Spatial	
	Easy	Mixed	Easy	Mixed	Easy	Mixed
0	31.3	31.3	29.4	29.4	26.1	26.1
100	21.9	44.2	33.3	29.1	49.9	56.6
200	40.0	43.4	32.9	32.7	56.7	54.3
500	44.2	35.5	36.5	34.0	53.1	55.7

Table 2. Mean test accuracy (%) vs training sample size across all datasets and difficulty settings. First row shows base model performance (Qwen3-4B with no fine-tuning).

that reinforcement learning was more stable and effective on larger, easier datasets in this setting.

Mixed-difficulty training was more challenging. Validation rewards were frequently negative across all dataset scales (Figure 2b), reflecting incomplete rollouts where the model exhausted its token budget before producing a parseable structured output. On average, mixed datasets contained larger graphs (14.9 nodes vs. 12.6 for easy sets), resulting in longer input and output sequences that more often exceeded the maximum generation length. Since the reward function penalized cutoffs (Section 3.3), these incomplete rollouts suppressed training rewards and slowed convergence. The reward component breakdown (Figure 5) confirms that extraction failures account for the majority of completions across all graph configurations (59–65% for easy, 67–73% for mixed), explaining the persistently negative aggregate rewards. Notably, the *Mixed-100* model performed slightly worse than the baseline (29.1% vs. 29.4%), coinciding with limited diversity and frequent negative reward signals from incomplete rollouts.

Test-Time Generalization. Easy-trained models slightly outperformed mixed-trained models overall at test time (Figure 3b). However, both training setups failed to generalize beyond the easy test subset. Performance declined sharply on medium problems and became mostly negative on hard problems, indicating that the models struggled to reason

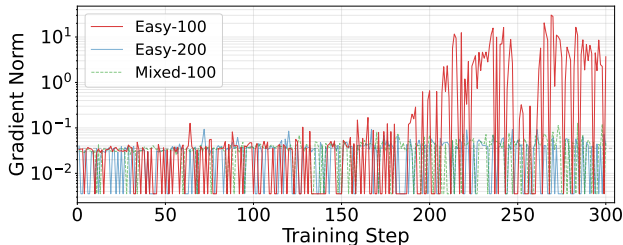


Figure 4. Gradient norm over training steps for three counting configurations. Easy-100 exhibits spikes exceeding $850\times$ baseline between steps 150–300, coinciding with the reward collapse in Figure 2a. Easy-200 and Mixed-100 remain stable throughout, supporting a minimum diversity threshold for stable optimization.

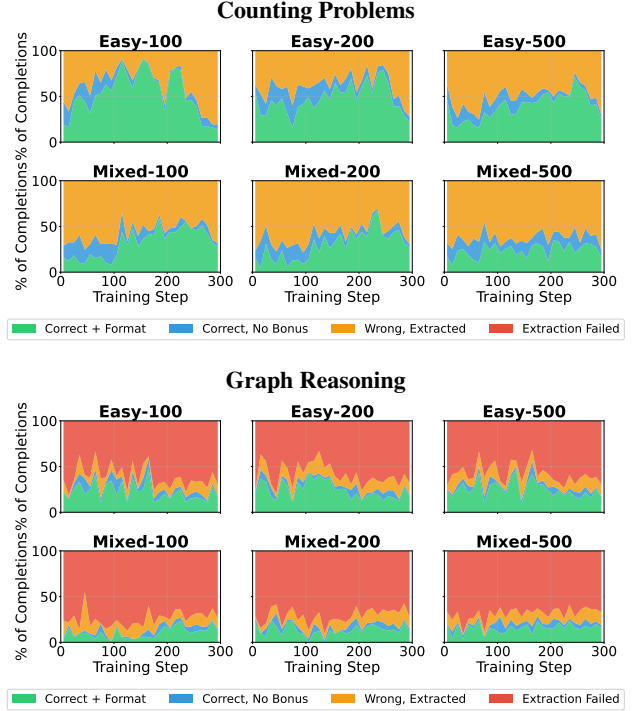


Figure 5. Reward component breakdown across training configurations. For counting, Easy-100 correctness collapses after step 150, consistent with gradient norm instability (Figure 4). For graph reasoning, extraction failures dominate (59–65% easy, 67–73% mixed), explaining persistently negative aggregate rewards. Spatial reasoning is excluded (binary reward, no sub-components).

over longer or more complex graphs. This is consistent with training inefficiencies under compute and context constraints, which limited the model’s ability to complete reasoning and reduced opportunities to receive positive reward signals on more difficult rollouts.

Scaling analysis across 100–500 examples revealed only incremental improvements within each training regime (Figure 2b). Easy-trained models scaled roughly linearly in final validation reward, while mixed-trained models showed minimal change. This weak scaling trend supports the conclusion that *training on harder instances—which involve longer sequences and more reasoning steps—is constrained by token limits and fixed compute budgets.*

Overall, graph reasoning scaling was constrained more by token generation limits than by data volume, given the verbose nature of the problems. Larger graphs inflated input and output lengths, causing the model to more frequently exhaust its token budget before producing complete outputs. Increasing these limits would change the underlying training dynamics and may yield different scaling behavior, making length-adaptive optimization an important direction for future work.

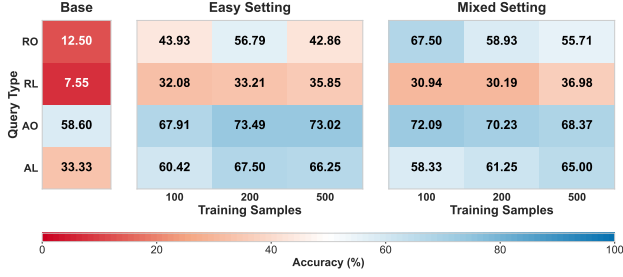


Figure 6. Test accuracy on different types of queries in spatial reasoning. Here, AO, AL stand for absolute orientation and absolute location, and similarly, RO, RL stand for relative queries. In both easy and mixed settings, we see improvements across all types of queries. The improvements are more pronounced on location and relative orientation queries. Moreover, in the mixed setting, the performance is much better on the relative orientation queries.

4.1.3 Spatial Reasoning

Training-Time Observations. During training, we see that the validation accuracy improves gradually in all settings (Figure 2c). The rate of improvement varies across the settings—the model learns faster in easy settings with fewer samples in comparison to other settings with more samples, especially with medium and hard difficulty levels. This is expected behavior with a fixed training budget across these settings; medium and hard questions are expected to be harder to learn and may require a longer training time and larger token limits.

Test-Time Generalization. After fine-tuning the model, we evaluate it on a test set of 200 problems of mixed difficulty levels (68 easy, 66 medium, and 66 hard) spanning 4 query types: absolute location (48 questions), absolute orientation (43), relative location (53), and relative orientation (56). We draw the following insights from the results on the test data.

Fine-tuning with either type of training set improves accuracy. While this is expected from fine-tuning in general, as a first step, it is necessary to study whether fine-tuning is effective in this novel setting of spatial reasoning. These results (in Table 2) provide clear evidence that fine-tuning improves the model’s performance (up to $2\times$) on the given spatial reasoning task.

Diminishing returns with more training data under compute constraints. Contrary to conventional wisdom, more training data may not yield higher accuracy when the compute budget for training is fixed. Our results on spatial reasoning in both easy and mixed training settings in Table 2 support this claim. We see that in the easy setting the accuracy improves by about 7% when moving from 100 to 200 training samples, but drops by 3.6% in the 500 samples case. In the mixed setting, the accuracy does not change much with the number of samples, but we see a small dip when compared

to the 100 samples setting. These results suggest an inverted U-shape scaling of test accuracy with the training set size under compute constraints.

Training on mixed difficulty samples generally performs better in comparison to training only on easy samples. In Table 2, we observe that training on samples with mixed difficulty results in a higher or comparable test accuracy, in contrast to training on an easy set of the same size. Moreover, the accuracy with 100 mixed samples even surpasses the accuracy with 500 easy samples, suggesting training on fewer samples of varying difficulty is preferable to training on a large set of easy samples.

Improvements across all types of queries. Next, we study the performance of the model on different types of queries before and after fine-tuning. The results are shown in Figure 6. We observe that fine-tuning with either training set improves accuracy across all query types. The baseline results suggest that relative queries are harder, and it is interesting to see that even training on easy samples increases the accuracy on these queries significantly. Similar improvements are also noted in the mixed setting, which are more pronounced on the relative orientation queries. This may reflect the fact that the mixed set contains more such queries in comparison to the easy set.

Robustness considerations. Due to computational cost, we did not perform multi-seed repetitions. We instead rely on consistency of qualitative trends across 18 configurations ($3 \text{ domains} \times 6 \text{ data configurations}$) as a robustness signal, suggesting the observed effects are not solely driven by training noise. The mixed-vs-easy sample efficiency trend emerges independently in both Counting and Spatial Reasoning despite differing reward structures, and all configurations share fixed training budgets, token limits, and task-specific reward functions, so performance differences reflect how dataset size and composition interact with these constraints. We view these findings as empirical observations under low data and compute regimes that generate hypotheses rather than definitive causal claims.

4.2 Implications

Our results across the datasets point to a unifying theme: under fixed-budget RLVR regimes, performance is shaped by interactions between dataset size, composition, training duration, and token limits. We distill three design lessons for practitioners.

1. Training-set composition can outweigh data volume. In both Counting and Spatial Reasoning, small mixed-difficulty datasets matched or exceeded the test accuracy of larger easy-only datasets. For Counting, 100 mixed examples matched 500 easy examples in test accuracy, a $5\times$ **sample efficiency advantage**, suggesting that curating across

empirically defined difficulty tiers can be more effective than scaling up easy examples alone.

2. Under fixed training budgets, scaling data alone can be ineffective. Larger datasets receive fewer optimization updates per example under a fixed step budget. In Counting, mixed-difficulty test accuracy declined beyond 100 examples even though validation rewards were still improving at the final step. In Spatial Reasoning, easy-trained accuracy peaked at 200 examples before declining, and mixed-trained accuracy did not improve beyond 100 examples. These patterns motivate joint consideration of dataset size, training duration, and token budgets.

3. Harder instances can be constrained by incomplete rollouts under fixed token limits. In Graph Reasoning, larger graphs frequently exhausted the token budget before generating parseable outputs, suppressing positive reward signals (Figure 5). Counting and Spatial Reasoning, with shorter outputs, were less affected. For domains requiring verbose reasoning, token budget allocation may be a more binding constraint than data size.

Consistent with the multi-factor framing in Section 3.1, these findings reflect interacting system constraints rather than effects of a single “complexity” factor. Alternative contributors, including optimization dynamics, reward sparsity, and task-specific reward design, may also shape the observed patterns.

5 CONCLUSION

In this work, we have presented a systematic characterization of open-source SLMs fine-tuned using RLVR under low data regimes. Using three procedurally generated datasets with controllable properties (covering number counting problems, graph reasoning, and spatial reasoning), we characterize the effects of dataset size, diversity, and complexity on model reasoning capabilities. We find that even within low data regimes, under RLVR, open-source SLMs trained on low complexity tasks generalize to higher complexity tasks not seen during training, and training on a mixture of complexities is associated with greater accuracy gains under lower data budgets—specifically, we find that the mixed complexity setting provides up to $5\times$ the sample efficiency versus training on easy tasks alone. Our results also demonstrate that procedural data generation is a useful tool for starting to understand the data scaling laws that govern the effectiveness of RLVR, and these findings motivate wider research into the development of empirical laws that relate reasoning capabilities after fine-tuning to dataset composition and size.

Although our work begins to characterize the effectiveness of RLVR in low-data regimes, several limitations should be noted. Our findings are specific to a 4B parameter

model with LoRA fine-tuning under fixed low compute budgets, and we did not perform multi-seed repetitions. These results are intended to inform practitioners operating under similar constraints, not to claim universal scaling laws. While data composition effects may remain relevant at larger scales (Tan et al., 2025), the quantitative gains (e.g., $5\times$ sample efficiency) may not directly transfer to larger models. Additionally, procedural tasks do not capture the full complexity of real-world data, and we did not evaluate transfer to natural language benchmarks.

Future work could develop more extensive scaling laws relating dataset properties to post-fine-tuning performance, validate these trends at larger model scales and compute budgets, evaluate transfer to natural language benchmarks, and develop formal budget-aware RLVR theory that captures interactions between optimization budget, token limits, and reward sparsity. As shown by Khatri et al. (2025), this requires significant computational resources.

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